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DEFINING THE CLINICAL COMPETENCY STANDARD AT THE UNIVERSITY OF PAVIA – A PILOT STUDY ON THE HARVEY COURSE

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1. Introduction

1.1 The Three Pillars of Modern Medical Education

“Medical Education” is a wide term, a system spanning undergraduate, postgraduate and continuing medical education. Though, initially, any attempt at defining it seems almost impossible, like most complex systems, it can and should be broken down into simpler, more digestible components. Modern pedagogy usually divides medical education into 3 foundational pillars, on which the rest of the structure stands: Knowledge, Skill, and Attitude[1–4].

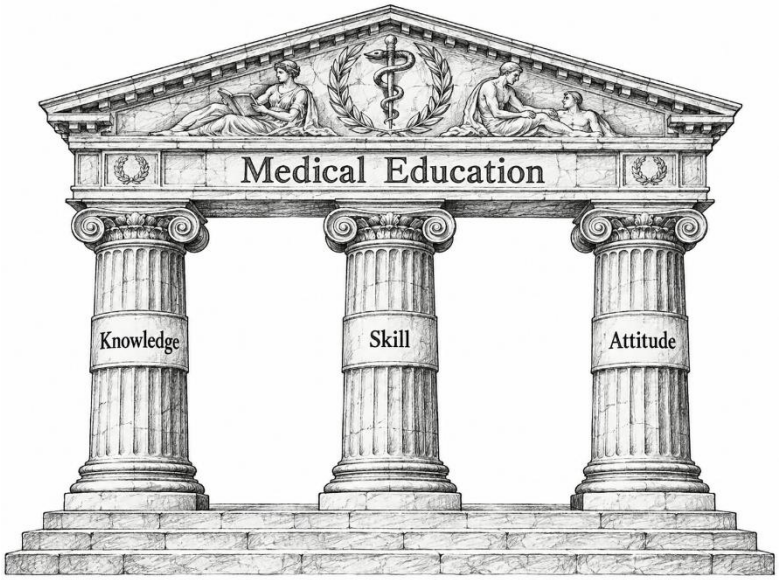


Figure 1 - Illustration summarizing the three pillars of Medical Education: Knowledge, Skill, and Attitude.

Once broken down into these 3 separate, but interdependent tracks, discerning the end goal and the process of medical education becomes much easier, and so does formulating its definition. As the World Medical Association defines it, basic medical education should "ensure that medical students have acquired the knowledge, skills, and professional behaviors" they need for practice[2].

Each pillar has an intuitive core.

- Knowledge is the foundational, factual, and mechanical understanding of disease, physiology, pathology, and treatment, built mostly in the classroom and through independent study.
- Skill is the ability to translate this knowledge into action, by physical examination, a procedure, or a therapeutic maneuver.
- Attitude is the professional disposition that mandates how knowledge and skill be used when facing patients and colleagues.

The three are dependent on each other[3,5], as skill without knowledge can be dangerous and misguided, and knowledge without skill is of little use at the bedside: both are not safe without the attitude to apply it responsibly. While they can be taught separately, this practice can easily lead to an imbalance, which in turn leads to a physician who is stunted in one or more domains.

Medical teaching has been reformed several times since the unstandardized apprenticeship model, first with the move to scientific, university-based training[6] and later with the move to evidence-based medicine[7]. The need for small-group, supervised bedside teaching has not changed throughout these reforms, and it remains one of the most important methods for teaching students the skills of the profession[8], and, arguably, the need for this intimate bedside teaching even grows in the age of expanding large classes[9–12].

Scale is the main stressor on the integrity of the three pillars. As medical programs grow in response to the increasing demand for doctors, the number of students per teacher increases, both in the classroom and, even more importantly, in the clinic and hospital corridors[12]. The effect is felt the most at the bedside, both by students and supervisors; about fifty years ago,

roughly three-quarters of clinical teaching took place there, while current estimates put the share at less than a quarter[8,13].

Out of the three pillars, knowledge has coped with this best: it was always suited to the classroom, and technology has put it more directly in students' hands than ever. Attitude also scales reasonably well, since it grows through a mix of clinical exposure and structured discussion. Skill is the most exposed of the three because it needs something the other two do not: repeated, supervised, hands-on practice. That is the part of training where a widening teacher-to-student ratio hurts most.

This thesis focuses on the skill pillar, and more specifically on the technical clinical skills a student should have by graduation. Before going further, we need a clearer definition of what "skill" means in a clinical setting, and where our scope begins and ends within it.

1.2 Defining "Skill": Scope of This Thesis

A clinical skill has been defined by the Task Force on the Clinical Skills Education of Medical Students as "any discrete and observable act of medical care"[14]. While this is a clean definition, for our purposes, the "acts of medical care" have to be taken apart and described in full. A 2012 international Delphi consensus addressed this by grouping clinical skills into four domains: physical examination, practical procedures, communication, and treatment or therapeutic skills, with near-unanimous agreement on each[15].

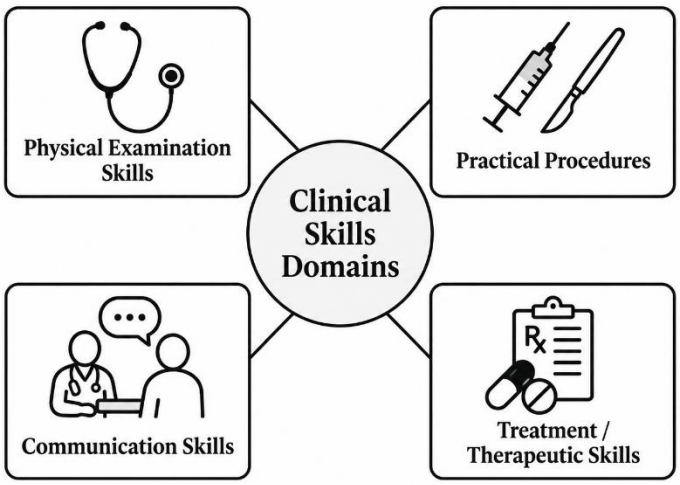


Figure 2 - The four domains of clinical skills as defined by Michels et al.[15]

These four domains can also be viewed through a second lens, as technical or non-technical, according to the faculties they draw on, rather than the task performed. Technical skills have been described as psychomotor actions, together with the related cognitive processes, acquired through practice and learning within a particular profession[16]. In medicine, these commonly include the cognitive, motor, and dexterity skills involved in physical examination, clinical procedures, use of medical equipment, and administration of medicines[17]. Non-technical skills, in contrast, are the cognitive, social, and personal abilities that support technical performance, including situational awareness, decision-making, teamwork, leadership, and communication[18,19]. Within this framework, physical examination and practical procedures are predominantly technical, while communication is non-technical; therapeutic skills may involve both technical and non-technical components depending on the task.

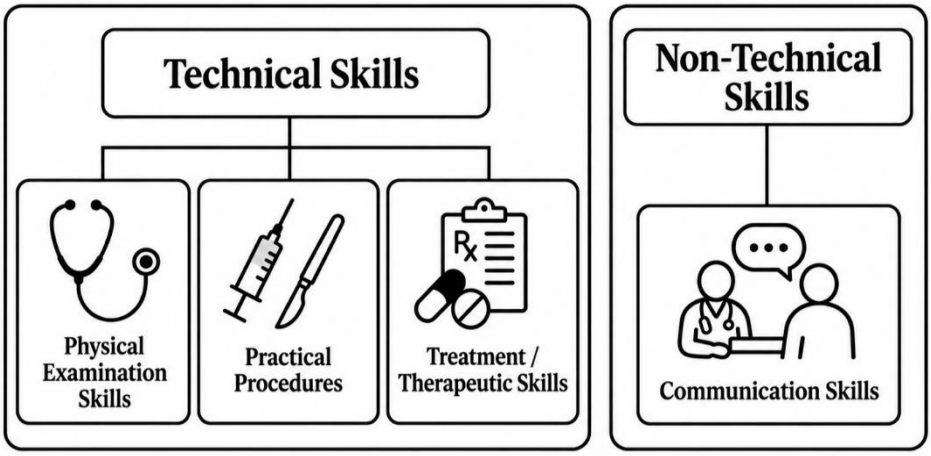


Figure 3 - Technical vs non-technical skills in medicine Agha RA et al.[16]

This thesis deals with technical clinical skills only: the psychomotor, hands-on part of clinical practice, covering examination, procedures, and therapeutic maneuvers. Communication and the other non-technical domains belong to the skill pillar just as much, but they fall outside the scope of this project.

Section §International **Approaches to Defining Clinical Skills** returns to this division, since several international frameworks draw the same line in their own skill lists.

1.3 Assessing Clinical Competence

With the term defined and the scope set, the next question is how to assess competence in a clinical skill: *How do we know when a student has actually acquired one?* A student who describes a procedure correctly in an exam has not necessarily performed it; a student who has performed it once under close supervision has not necessarily mastered it well enough to be trusted alone. Medical education has developed several frameworks to close the gap between describing competence and verifying it, and this thesis relies mainly on two: Miller's Pyramid[20], which describes the levels through which clinical competence is built and assessed and the model of

Entrustable Professional Activities[21] which turns that pyramid into a practical decision: whether a given learner can be trusted with a given task, and how much supervision they still need.

1.3.1 Miller's Pyramid

The most widely used model for structuring the assessment of clinical competence is the pyramid proposed by George Miller in 1990[20]. It has four levels:

1. Knows: the learner can recall and explain the relevant information.
2. Knows how: the learner can apply that knowledge to a clinical problem, usually tested through written or oral scenarios.
3. Shows how: the learner can perform the skill under observation, typically in a controlled setting such as simulation, but not yet in real practice.
4. Does: the learner performs the skill in actual clinical practice, unprompted and unobserved.

The value of the pyramid lies in the distinction between its levels rather than the labels it assigns them. Most traditional assessments in medicine, written and oral exams alike, only explore the bottom two levels: they confirm that a student is aware of a procedure, its indications and, sometimes, their understanding of the procedure; it says nothing about whether the student can perform it, how reliably, and under what supervision or independence level. By the definition adopted in §Defining **"Skill": Scope of This Thesis**, a skill sits at the "shows how" and "does" levels, which is where knowledge-oriented assessment is weakest. As later chapters argue, it is also where the emphasis of the Italian Core Curriculum may fall short.

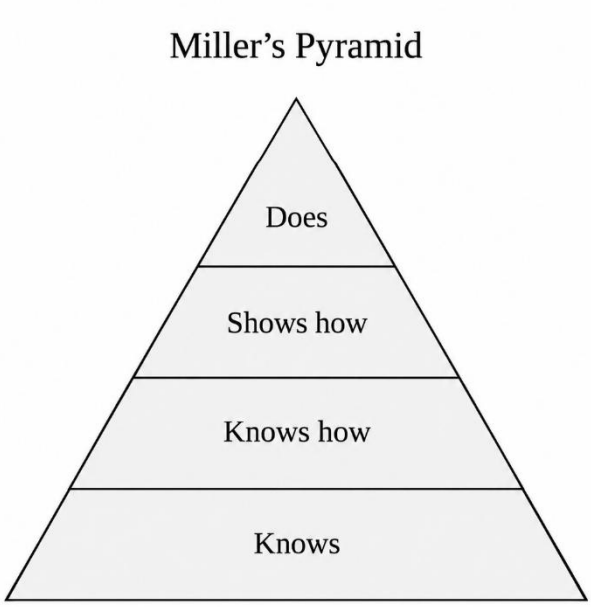


Figure 4 - The 4 levels of Miller's Pyramid[20].

The same distinction that makes the pyramid of value to us is also what makes it difficult to apply: "Shows how" can be observed directly, at simulation sessions, during a supervised procedure, or with an examiner's checklist; 'Does' cannot, because this level of independence removes the supervisor checkpoint from the equation. A curriculum that only defines what a graduate should know, or know how to do, never actually attempts achieving the competency of the top two levels. Without that commitment, a statement such as "the student should be able to perform basic procedures" is an intention, rather than an educational goal as it does not set any level of independence.

Closing this gap requires a different question: instead of asking whether a student can demonstrate a skill on request, we ask whether a student can be trusted with this task, and with how much supervision. The model of Entrustable Professional Activities was designed around that question[21].

1.3.2 Entrustable Professional Activities (EPAs)

The term Entrustable Professional Activity (EPA) was introduced in 2005 by Prof. Olle ten Cate. An EPA is a "task or responsibility to be entrusted to a trainee once sufficient specific competence is reached to allow for unsupervised practice"[21]. This gives the supervisor something more tangible to judge: whether the student can be trusted to carry out a specific task without direct supervision, as opposed to whether the student knows the relevant material. Trusting students with health care tasks "means more than observing adequate knowledge and skills as examined in tests"[22]: a test result shows what a learner knows or can demonstrate when asked; an entrustment decision is a supervisor's judgment, built on repeated observation, that a particular student can be allowed to act with a certain amount of supervision – or the lack thereof – in a particular context.

This standard is more specific than the jump from Miller's "show how" to "Does", and provides a more controlled mechanism of verification, moving from a simulation or highly controlled situation into the real clinical world.

Unsupervised competence becomes a staged target that can be tracked over time: each activity is entrusted step by step as competence grows, and the trainee moves from full supervision toward independent practice[21]. Ten Cate et al. later formalized this progression as a five-level supervision scale, from Level 1, where the trainee only observes, to Level 5, where the trainee may supervise others in the same activity[23]. The scale gives curriculum designers a shared language for describing how much autonomy a learner has earned at a given point.

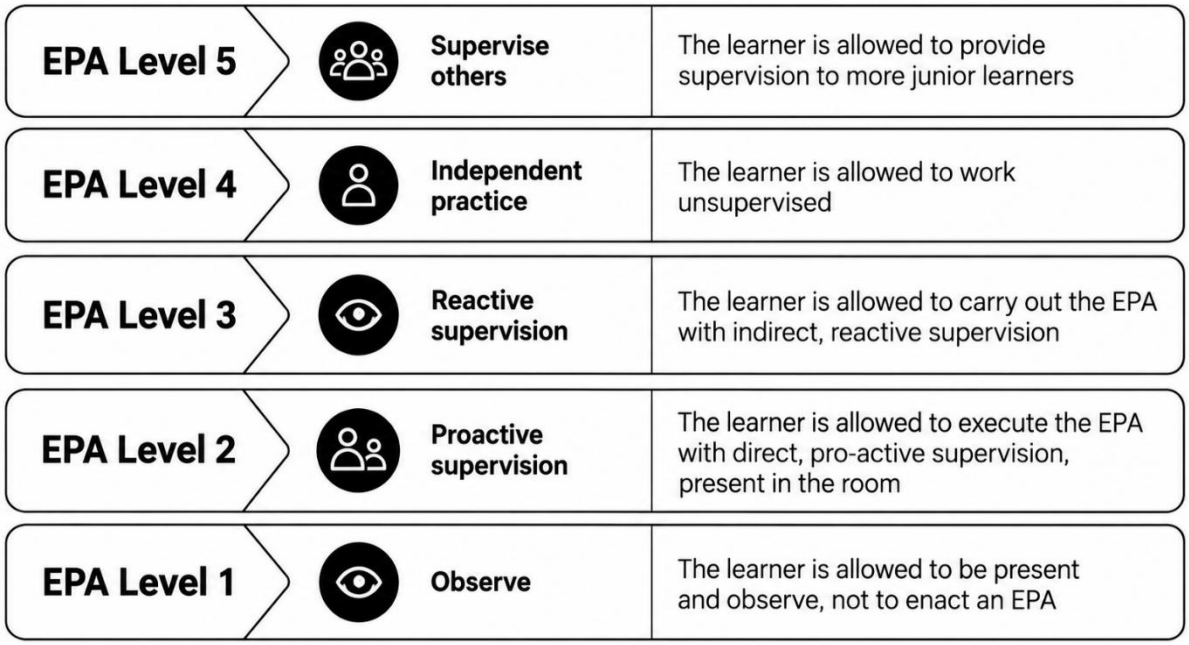


Figure 5 - The Entrustable Professional Activity (EPA) scale[23–26].

Adapted to the technical skills defined in §Defining **"Skill": Scope of This Thesis**, this scale is the basis of the survey instruments used in this thesis. The design follows an existing precedent: a Swiss study used a similar entrustment-supervision scale to compare a curriculum-defined expected level of competence with graduates' self-reported level across 46 EPAs, and treated the gap between the two as a baseline for curriculum reform[27]. We apply the same logic in a setting where no catalogue of expected levels exists. In Switzerland, PROFILES provided the national standard to compare to. At the University of Pavia, we first have to build that standard through a faculty expectation survey, and only then measure student self-assessment against it on the same scale. The gap between the two serves as a proxy for the distance between current training and current expectations.

1.4 Competency-Based Medical Education (CBME)

Miller's Pyramid and the EPA model belong to a broader change in how medical training is organized, known as competency-based medical

education (CBME). CBME can be defined as "an outcomes-based approach to the design, implementation, assessment, and evaluation of medical education programs, using an organizing framework of competencies"[28]; it can be contrasted with time-based training, still the dominant model in much of undergraduate medical education. In a time-based program, students' progress by completing a set number of years, courses, or hours, on the assumption that competence follows from time served; CBME reverses this: progression thus depends on demonstrated ability, and a learner moves on once a defined competency has been shown[29].

This transition is more than an administrative change. CBME rests on three things: a clearly stated outcome for what a learner must be able to do, training that moves through a planned sequence at a pace suited to the individual, and competence checked through entrustment and real workplace assessment rather than attendance or exam scores[29,30]. Miller's Pyramid maps what needs to be assessed at each stage, and the EPA model provides a way to make that assessment concrete. CBME requires both, so a curriculum cannot claim success simply because students have attended the required hours of teaching.

This integration has a further benefit that links back to §The **Three Pillars of Modern Medical Education**: practicing a technical skill is rarely a purely psychomotor act, and teaching a skill together with the knowledge behind it improves the performance of the skill itself. In a randomized trial on lumbar puncture, medical students who were taught the anatomical rationale alongside the step-by-step technique retained and transferred the skill significantly better than students taught the procedure alone, and the improvement in conceptual understanding mediated the improvement in hands-on performance[31]. Practicing a skill also places the learner, even

briefly, in the clinical situation the skill is meant for: a real or simulated patient, real stakes, and a real need for the composure, communication, and judgment that make up the attitude component of medical education. Skill practice therefore develops alongside the other two pillars, and, when done well, it exercises all three at once in a way lectures and written exams cannot. For the same reason, a curriculum that neglects the skill pillar is unlikely to make up for it through strength in the other two. Knowledge and attitude also need structured chances to be used in context, and, as the lumbar puncture trial suggests, skill practice is one of the most reliable ways to create such opportunities within a training program.

CBME and the three-pillar model therefore work together: CBME provides the structure through which skill, once defined and made measurable by Miller's Pyramid and the EPA model, can be built into a curriculum, sequenced and tracked as a required part of the course, instead of being left to whichever exposure a student happens to receive.

1.5 The Reference Point: Graduation vs. Transition to Residency

Any statement about educational goals or what a graduate "should be able to do" needs a discrete point of reference. There are three possible anchors: the last day of medical school, the transition to residency (a period, rather than a single moment), or the first day of residency. The choice matters: depending on the system, these moments can be days or years apart, and a skill list designed for one does not automatically apply to another.

Each regional framework addresses this issue differently, typically reflecting the practices and boundaries specific to that point in its system.

The UK's General Medical Council ties its standard to the moment a graduate begins working professionally[32]. However, in the UK, the

transition into work comes with a formal safety net. Newly qualified doctors spend a full Foundation Year (F1) under structured supervision, with provisional rather than full registration, before they are allowed to work independently[33]. For practical purposes, graduation and the first day of supervised practice coincide, while the first day of actual residency or practice is buffered by a two-year period aimed at accomplishing the transition into the ‘Shows how’ and ‘Does’ of Miller’s pyramid.

Switzerland's PROFILES framework names the first day of residency explicitly as its target[27]. In Switzerland, that day comes only after a mandatory national licensing examination, and independent practice remains legally barred until a specialist examination at the end of postgraduate training, years later[34]. While the UK protects its graduates with one mandatory checkpoint before they act alone, Switzerland protects them with two.

Unlike the prior examples, Italy has none: though previously, a 3-month practical rotation[35] followed by a licensing exam was required to practice medicine[36], since the 2020 reform, the Italian medical degree has been ‘abilitante' (qualifying, licensing): graduation itself grants the legal right to practice, with no licensing exam and no compulsory internship between the degree and the patient[37]. A graduate can legally work alone immediately after graduating, in emergency shifts, locum posts, or private practice. The national residency entrance exam (SSM) takes place once a year in July[38], with students on the “usual” path sitting it between one week and nine months after graduation, but it is not a requirement for practicing in the meantime; residency itself is optional. A graduate may never sit the SSM and continue to practice indefinitely without any specialty training. In summary, while the UK adds two supervised years of training after

graduation and Switzerland adds a licensing exam and years of supervised specialist training, Italy adds nothing.

This concept is what determines our reference point. In the UK and Switzerland, the first day of residency and the last day of medical school are meaningfully different anchors, because a legally enforced interval of supervision, examination, or both, separates them. In Italy, instead, these days coincide: graduation is the first day on which a doctor may be asked to act entirely alone. Using the first day of residency as the reference point would impose a checkpoint that is not recognized within the Italian system, and that would thus measure students against a level of competence that they are not expected to demonstrate before undertaking unsupervised practice. We therefore anchor this thesis to the day of graduation, meaning the newly graduated doctor on the first day of independent practice. The survey instruments we use and are described in later chapters use the same framing: in practice, this means assessing students at graduation or as they approach it, including sixth-year students close to the end of the course. In the Italian context, this is the same standard that "first day of residency" represents elsewhere, because it is the point at which unsupervised practice can begin.

Whichever moment is chosen, the logic of §Assessing **Clinical Competence** still applies: the target has to be a concrete, assessable claim at the "does" level of Miller's Pyramid, expressed as the level of supervision a specific learner has earned, and not a vague statement about what a well-trained graduate should in principle know how to do.

1.6 International Approaches to Defining Clinical Skills

Once the reference point is fixed, the next question is what exactly a graduate should be trusted to do by then. Different frameworks answer this in noticeably different ways.

Switzerland's PROFILES framework is a national catalogue for undergraduate medical education with three parts: general competencies adapted from CanMEDS, a short list of core entrustable professional activities, and a large set of clinical situations in which those activities are performed[39]. Expectations are stated as activities, and one common target level applies across the core EPAs, described as supervision from a distance and on demand[27,39].

The GMC takes a different approach: Outcomes for Graduates organizes its expectations into three broad areas: professional values and behaviors, professional skills, and professional knowledge[32]. A separate document, Practical Skills and Procedures, lists the specific practical procedures a newly qualified doctor must be able to carry out safely[40]. Keeping the technical list apart from the general framework treats procedural skill as its own category, named separately, which is similar to the line drawn in §Defining **"Skill": Scope of This Thesis**.

At a supranational level, the Tuning Project sought to address the issue on an even broader scale: a shared reference framework for medical degree programs across the European Union[41]. In Diogo et al.'s implementation of Tuning in Portugal (the TEST study), the Clinical Practice domain was large: 12 Level 1 competences and 70 Level 2 items[42]. These items place practical procedures alongside communication, ethics, evidence-based practice, and population health[41,42]. Some individual procedures are named

at the second level, but with no level of independence attached[41,42]. Tuning therefore sits closer to the GMC's broad first tier than to a targeted skill list, and it does not offer the itemized, level-graded second tier that would make it directly comparable to Practical Skills and Procedures.

When analyzed together, these examples show real variation in how "the same" target gets defined: an activity-based national framework with one common target level (Switzerland), a two-tier structure separating broad domains from an itemized skill list (UK), and a supranational competence framework that bundles technical skill with several other domains (Europe, through Tuning).

1.7 The Italian Context

1.7.1 Absence of a National Technical Skills Consensus

Applying §International **Approaches to Defining Clinical Skills** to Italy shows the same absence found across every model reviewed there. Italy has no equivalent of Switzerland's PROFILES, the GMC's Practical Skills and Procedures, or even the broader domain structure Tuning provides. The national Core Curriculum developed by the Conferenza Permanente, built around roughly 1,659 teaching units (UDE) in its 2017 revision[43,44], remains largely knowledge-based: about 73% of the units are classed as theoretical and about 18% as practical, and only 14 units (0.8%) reach the level of autonomous ability[44]. Before that revision, a member of the commission that built it described the prevailing units as "conoscenze da memorizzare più che competenze e abilità", knowledge to memorize rather than competencies and skills[45]. The practical units are a minority, scattered through a mostly theoretical list, and they are not set out as a list of technical skills with a required level of independence for each.

This is not to say the ministerial decrees are entirely silent on practical competence. Decreto Ministeriale 58/2018 defines the Tirocini Pratico-Valutativi described in §Why **the Absence Matters More in Italy** explicitly as verifying a student's "saper fare e al saper essere medico," knowing how to do and knowing how to be a doctor[36,46,47]. In practice, however, the assessment is holistic rather than itemized: a single libretto-diario, completed by one tutor, concluding in a simple pass or fail, idoneo or non idoneo (eligible or non-eligible), for each rotation. No skill-by-skill checklist specifies which procedures a student must demonstrate, or at what level of independence, before that judgment is given. The body responsible for Italian medical curricula has itself acknowledged this gap. Writing shortly after the TPV became the sole route to licensure, the Conferenza Permanente called for greater attention to how these rotations are evaluated, pointing specifically to structured international methods, OSCE stations, portfolio and workplace-based assessment, the Mini-CEX, direct observation of procedural skills, as approaches that would suit this kind of judgment well[46,48]. The decree establishes that practical competence should be assessed; it does not yet establish what, specifically, must be shown, or how.

1.7.2 Why the Absence Matters More in Italy

In any system, the absence of an official, well-defined list of practical and technical abilities, a clear clinical standard for what a graduating student must reach, would be a shortcoming of the curriculum. In Italy, the consequences go further.

As established in §The **Reference Point: Graduation vs. Transition to Residency**, Italy's 2020 reform removed every checkpoint between graduation and independent practice, thus resulting in: no licensing exam, no compulsory

internship following graduation, and a specialty exam that is voluntary and can be delayed indefinitely or skipped altogether. Where the UK and Switzerland each interpose at least one mandatory stage between graduation and full independence, Italy interposes none.

As a result, undergraduate training is not always a step toward more structured clinical education; for some Italian graduates, instead, this represents the only structured clinical training they will ever receive. Where postgraduate training is mandatory, an undergraduate skill gap can still be identified and corrected at the next stage; however, in Italy, a graduate who does not specialize has no next stage. A defined, standardized level of clinical skill at graduation is therefore the only way to make sure that every graduate who starts practicing independently the day after graduation has first reached a demonstrable standard.

1.7.3 The Gap Persists into Residency

For graduates who do specialize, the gap does not close by itself; Italian postgraduate training is fragmented as well. A recent paper from Pavia's Emergency Medicine group describes marked heterogeneity across Italy's 36 emergency medicine residency programs in structure, clinical rotations, and access to simulation-based training, and notes that no nationally codified simulation curriculum exists[49]. The lack of a defined skill standard is therefore not always corrected once graduates enter postgraduate training; the same structural gap may reappear one step later, especially in newer specialization training programs, such as Emergency Medicine[50].

1.8 The Local Context: University of Pavia

The University of Pavia offers its single-cycle degree in Medicine and Surgery (LM-41) in two parallel courses: Golgi, taught in Italian, and

Harvey, taught in English. Golgi is the older and larger of the two. Harvey started in 2009 as the first medical course in Italy taught in English[51]. It follows an Anglo-Saxon-style curriculum, currently has 143 places per year, and draws more than 40% of its enrolled students from abroad[51]. Both courses award the same degree with the same legal effect described in §Why **the Absence Matters More in Italy**: an abilitante qualification that allows graduates to practice from the day they graduate.

Both courses also describe the doctor they aim to produce in similar terms: a physician with the scientific foundations, theoretical and practical preparation, and professional competencies to work in positions of responsibility, in Italy and abroad. Harvey's objectives state explicitly that they are built on the WFME and TUNING-CALOHEE standards[52]. On a local scale, there may therefore be an opportunity to reduce the gap previously described in §Absence **of a National Technical Skills Consensus**. While the course documents cite international, competency-based standards as their foundation, this alignment is not yet fully translated into a clearly defined set of technical skills that a graduate is expected to demonstrate and that can be systematically assessed.

The University of Pavia is well-equipped and currently has a strong basis for closing this gap, because much of the infrastructure it needs already exists. The Clinical Skills Lab, directed by Professor Andrea Pietrabissa, provides a dedicated space for structured skills teaching. A collaboration with Lab3D4Med has produced 3D-printed skill trainers, made in house, for procedures ranging from suturing to ultrasound-guided percutaneous analgesia. Simulation facilitators trained through SIMMED, and in some cases through the European EEDUSIM program, already run structured simulation teaching at Pavia[49]. Skill transfer between students is also

already happening informally, alongside official channels: senior students tutor and mentor juniors, both through structured, university-recognized tutoring programs and through informal, peer-to-peer arrangements that students organize themselves. Local studies show this infrastructure producing results, including at the undergraduate level, demonstrating that simulation guided teaching carries over and improves students' clinical skills[53]. This is evidence from one pilot that structured skills teaching can work for the student population this thesis is concerned with. A later study applied a similar approach to training emergency physicians in ultrasound-guided regional anesthesia and, using a pre/post design, showed skill gains after simulation[54]. At the postgraduate level, Pavia's Emergency Medicine residency has developed a five-year, progressive simulation curriculum and mapped it against EUSEM and Italian standards[49]. Together, these studies show that the University of Pavia has the capability and the experience to design, deliver, and evaluate a structured skills curriculum, and the residency program shows that such a model can be developed within the same faculty.

1.9 Summary and Roadmap

This chapter moved from a general model of medical education to the specific problem this thesis addresses in Pavia. Medical training rests on three interdependent pillars: knowledge, skill, and attitude (§The **Three Pillars of Modern Medical Education**). We focus on skill, and, within it, on technical clinical skills: the psychomotor, hands-on part of practice, as opposed to communication and the other non-technical domains (§Defining **"Skill": Scope of This Thesis**). Written and oral exams cannot wholly demonstrate whether a skill has been acquired, as Miller's Pyramid separates knowing from doing and shows that traditional assessment rarely

reaches its upper levels (§Miller's **Pyramid**). The EPA framework addresses this by turning competence into a concrete entrustment decision that can be graded (§Entrustable **Professional Activities (EPAs)**). Competency-based medical education puts both to use: it ties progression to demonstrated ability rather than time served, and treats skill practice as an integral component able to strengthen knowledge and attitude instead of competing with them for teaching time (§Competency-Based **Medical Education (CBME)**).

Any such framework requires a clearly defined point at which competence is assessed. In other countries that point is usually followed by a period of protected, supervised practice before full independence, such as the UK Foundation Programme or the Swiss licensing exam and specialist training. Since the 2020 reform made the Italian degree abilitante, no such period exists in Italy, and graduation and unsupervised practice coincide. We therefore anchor this thesis to graduation, the point at which graduates face the greatest exposure to independent clinical responsibility within the Italian system (§The **Reference Point: Graduation vs. Transition to Residency**).

Other systems define graduates' technical skills in very different ways: Switzerland's activity-based framework with one common target level, the UK's two-tier structure separating broad domains from an itemized skill list, and the European Tuning framework's broader competence bundle, which folds technical skill in with several other domains (§International **Approaches to Defining Clinical Skills**). Italy matches none of these models, as its Core Curriculum remains knowledge-based, with no officially defined target for technical skills, and a similar fragmentation appears in postgraduate training (§The **Italian Context**). Pavia, however, already has much of what is needed to address the gap at undergraduate level; it

features a Clinical Skills Lab, 3D-printed trainers, trained simulation facilitators, local evidence that simulation improves student skills, and a five-year simulation curriculum developed in its own Emergency Medicine residency (§The **Local Context: University of Pavia**).

2. The Question

2.1 The Problem

The Harvey course has not yet formalized which technical skills a graduate is expected to master, or at what level of independence. The course's stated aim is to train physicians who can practice safely, professionally, and responsibly, with the course objectives built upon the WFME and TUNING-CALOHEE standards[52]. However, when it comes to translating this aim into a list of technical skills, actionable tasks, or determining the level of autonomy a graduate student should have when executing these skills, both the course documents and the national documents fall short (§Absence **of a National Technical Skills Consensus**).

Without a pre-defined standard, students may lack clear guidance on which clinical skills should form the core of their education and to what extent of independence they should strive at each stage of their training. Similarly, tutors may lack a common reference for the competence expected of a sixth-year student or for the skills to which students have previously been exposed. As shown in Appendix C: **Intra-Student Variability in Self-Rated Clinical Skill Readiness**, this mismatch can contribute to variability in students' clinical and technical experience by graduation.

As discussed in §The **Three Pillars of Modern Medical Education**, increasing student numbers place additional pressure on opportunities for supervised clinical practice[8,13]. In the absence of a structured clinical skills pathway, differences in exposure and progression may therefore be more difficult to identify and address. This thesis does not assume the magnitude of any resulting competency gap, but measures it.

The consequences of this gap were described in §Why **the Absence Matters More in Italy**. A newly graduated student can start practicing independently almost immediately after graduation, with no further checkpoints of training. For these graduates, the knowledge and skills acquired (or not acquired) during the course directly shape what they bring into clinical practice.

This situation parallels Flexner's historical report from 1910, which responded to an unregulated medical school system by pushing for a standardized curriculum[6]. This thesis addresses a narrower aspect of that broader argument, focusing on one course and one specific component of medical education - technical clinical skills. What a graduate can do should be pre-determined and achieved regardless of which ward or supervisors they were assigned to.

2.2 Research Question

This thesis asks one main question, divided into three parts:

1. **The standard.**
    a. Which technical clinical skills should form the core skill set expected of a Harvey graduate?
    b. What level of supervised independence should a graduating student be expected to reach in each of these skills?
2. **The gap.** How far do sixth-year students' self-perceived levels fall below those expectations, skill by skill?
3. **The response.** How could the identified gaps be addressed and minimized using current resources or feasible curricular changes?

The first part must be addressed and established before proceeding to the other two, as it provides the reference standard against which the gap can be assessed. The wording of the second part is deliberate: students report how competent they feel, which does not necessarily coincide with how competent they actually are. Accordingly, all findings in this thesis concerning the competency gap refer specifically to perceived competence, and are described as such throughout this work.

2.3 Aims and Objectives

This thesis aims to define the technical clinical skills expected of a graduating Harvey student, measure how far current students are from that expectation, and propose a way to minimize or close the gap.

This work follows four objectives, one for each phase of this study:

1. To map international frameworks' definition of graduate level technical skills, and the competency level they expect at graduation or an equivalent time frame (phase 1). This produces the list of skills and the supervision level expected globally, for comparison with the local standard we will establish in the next phase.
2. To establish a local faculty expectation of the level of competence and supervision for each skill (phase 2).
3. To measure students' self-prescribed level on the same items list, and then quantify the gap for each skill (phase 3). Using the same items and the same scale as Phase 2 allows a direct comparison without any required translation between the two.
4. To propose a thematic curriculum, built around a small set of guiding principles and within current logistical and administrative

constraints, with suggested ways of applying each principle to the skills where the gap is largest (Phase 4).

Objectives 1 to 3 are descriptive. Objective 4 is a proposal, based on the measured gaps, and it is designed to work within the resources and the logistical and administrative constraints that already exist in Pavia (§The **Local Context: University of Pavia**); however, it has not yet been piloted, and its effect on student competence has not been tested thus far.

2.4 Scope and Boundaries

Skills. Only technical clinical skills are included, as defined in §Defining **"Skill": Scope of This Thesis**. Clinical reasoning, communication, and attitude are outside the scope of this thesis; though they belong to the skill and attitude pillars described in §The **Three Pillars of Modern Medical Education**, and they have fundamental significance, they require different methods to define and assess.

Course. The study covers the Harvey course only. The Golgi course is not included at this stage. All Harvey students are treated as a single group, and no distinction is made between Italian and international students.

Population. The student sample is current sixth-year students, as they are the closest available group to the graduation reference point set in §The **Reference Point: Graduation vs. Transition to Residency**. Already graduated doctors are not sampled.

Measure. The gap is based on self-assessment, on both sides: faculty state an expected level, and students report their own perceived level. Self-assessed competence has been shown to agree poorly with observed performance[55], and, in at least one study of final-year medical students, the two were found to correlate negatively[56]. The gap reported here should

therefore be interpreted as a difference between expectation and perception, not as a measurement of what students can actually do. A comparable design, graduating students' self-assessed competence compared against their own teachers' expectations across 99 clinical skills, found the same direction of gap reported here, students rating themselves below what their teachers expected[57], which suggests this pattern is not specific to this course or country, though it does not establish that either side's rating reflects actual performance. Section §Limitations **of the Study** returns to this point.

Proposal. We offer 4 principles for building a clinical curriculum, that is well within the current logistical and administrative constraints. It sets out the principles and examples of how to apply them. It is not a full session-by-session design.

2.5 Expected Findings

This is a descriptive study, and it does not test a formal hypothesis. We expect faculty to set expectations close to the international average overall, with a lower bar for specialized or invasive procedures and a higher bar for foundational skills such as physical examination. We expect students to rate themselves below the faculty expectation on many skills, most clearly for invasive procedures. These are expectations only, and the results may show otherwise.

3. Methods

3.1 Study Design

The study has 4 phases, each one building upon the one preceding it to focus or question from a global perspective to a local one, all the while maintaining a practical scope.

First, we compared clinical curricula of national and supranational bodies and extracted common technical skills, together with the level of competence each expects at graduation (or its equivalent set point, as mentioned in §The **Reference Point: Graduation vs. Transition to Residency**): this is Phase 1.

Second, we have turned this list of skills into a survey and asked Faculty members of the Harvey course to rate the level of supervision they expect at graduation for each skill (phase 2).

Third, we gave the same survey to sixth-year Harvey students and asked them to rate their own self-perceived level of confidence currently (phase 3).

Fourth, we used the comparison between these scores to identify the largest gaps, and we built a curriculum proposal around them (Phase 4).

Therefore, each skill receives three scores: the international expected level, the level expected by the Harvey faculty, and the level reported by students. This comparison will allow us to recognize where the course falls short of the established expectations, and to quantify the extent of the identified gaps.

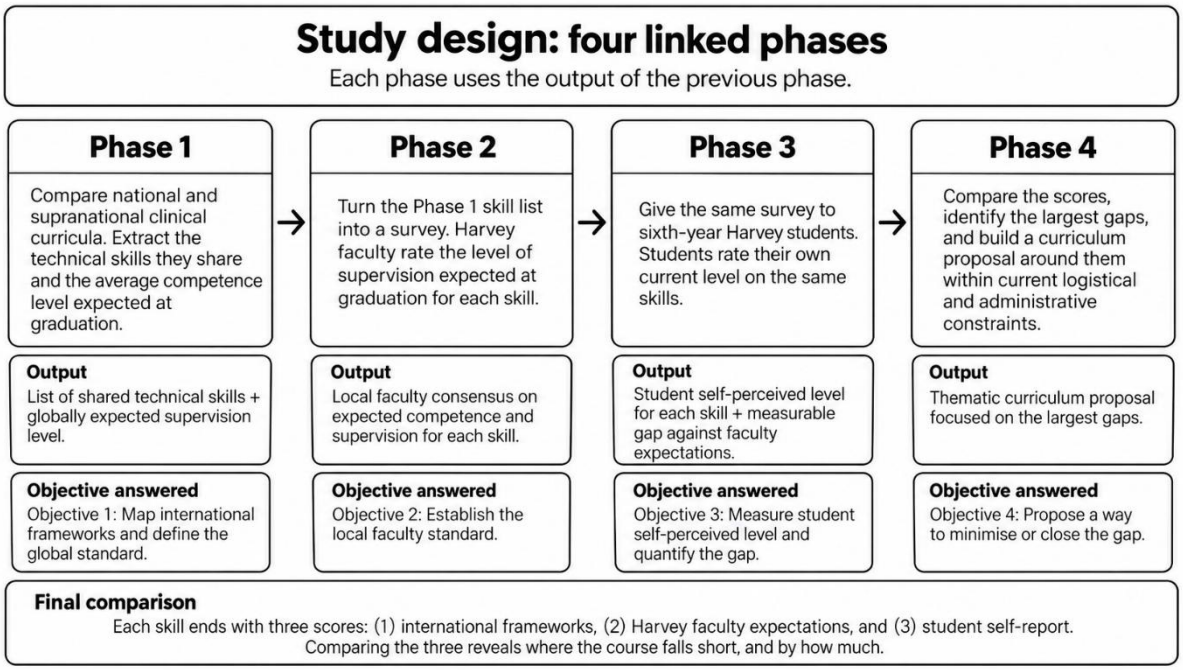


Figure 6 - Graphic depiction of the stages of the study design, identifying the main elements of each phase, their outputs and answered objectives. The graph aims at summarizing the evolution of this study throughout its implementation.

3.2 Phase 1: Comparison of International Frameworks

3.2.1 Sources

The skill list was built from three frameworks published at national level or above.

- **GMC (United Kingdom).** Practical skills and procedures (2019), which lists 23 procedures. Each has a minimum level of competence for a newly qualified doctor, on a ladder of three levels[40]. Outcomes for Graduates gives the wider framework[32].
- **PROFILES (Switzerland)**, 2017 edition. Nine core entrustable professional activities (EPAs). EPA 2 (physical and mental status) and EPA 5 (general procedures) contain the specific lists of examinations and procedures. One minimum level applies to all

EPAs: graduates must perform them at least under distant, on-demand supervision on the first day of residency[39].

- **Tuning Project for Medicine (Europe)**. Outcomes and competences for graduates, with no levels of supervision[41].

We did not use curricula published by single medical schools, because they report a single institution's choice rather than an agreed national or international position.

3.2.2 Conversion Rules

Every framework level was converted to the survey scale. The survey scale is: 0 - no understanding, 1 - observation only, 2 - active direct supervision, 3 - reactive direct supervision, 4 - indirect supervision, 5 - autonomous.

GMC

Descriptor	Survey Level	Reason
"Safe to practise in simulation"	2	The graduate has practiced only on simulators and needs direct supervision on the first real patients, which matches an instructor guiding every step.
"Safe to practise under direct supervision"	3	A supervisor observes the graduate performing the procedure, which matches passive supervision (the graduate performs and the supervisor intervenes only if necessary).
"Safe to practise under	4	The GMC defines this as being able to reach a colleague for help, which matches a

indirect supervision"		supervisor available nearby without direct observation.

Table 1 - GMC framework levels and descriptors.

PROFILES

Descriptor	Survey Level	Reason
Item in italics in EPA 2 or EPA 5 (demonstration only)	2	A demonstration on a simulator or model equals the GMC simulation level.
PROFILES: item in EPA 2 or EPA 5, or covered by another EPA, at the common level ("at least distant, on-demand supervision")	4	The supervisor is not observing and is called when needed, which matches indirect supervision.

Table 2 - PROFILES framework levels and descriptors.

Tuning

Tuning states outcomes without a supervision scale: it shows that a skill is listed and never sets a level.

No GMC descriptor and no PROFILES level reach 5 – as mentioned in Chapter Introduction –, as both frameworks set the base for continuous training and not for independent practice.

Coverage is recorded as Listed, Partial, or Not listed. Partial means the framework covers the skill only through a broader item, or covers only part of the survey item.

Reference level. The reference level of a skill is the highest converted level among the frameworks that list it, whether fully or in part. Tuning is not counted, because it has no levels. A skill listed by no framework has no reference level.

3.2.3 The Skill List

The full skill list including reference levels can be found in Appendix A: **Phase 1 Skill Synthesis — Full List**].

Basic monitoring and point-of-care testing

No.	Skill	What it means
1	Vital signs	Measure and record blood pressure, heart rate, respiratory rate, temperature, and oxygen saturation, and tell normal from abnormal values
2	Capillary blood glucose	Perform a fingerstick glucose test and interpret the result
3	Urinalysis, urine dipstick, pregnancy test	Perform and read a urine dipstick and a urinary pregnancy test, and interpret the results
4	Glasgow Coma Scale and ACVPU	Assess and score level of consciousness with the GCS and ACVPU in an acute setting
5	NEWS2 and early warning scores	Calculate an early warning score from vital signs and use it to recognize a deteriorating patient

Instrumental exams and diagnostic test performance (performing the test, not interpreting it)

6	Ophthalmoscopy	Use a direct ophthalmoscope to see the fundus and identify obvious abnormalities
7	Otoscopy	Use an otoscope to examine the external ear canal and the tympanic membrane
8	12-lead ECG	Place the electrodes correctly and record a technically adequate 12-lead ECG
9	Spirometry	Perform basic spirometry and guide the patient through the maneuver

Vascular access and sample collection

10	Venous access	Perform a venous blood draw, place a peripheral IV cannula, and obtain a blood culture with correct technique
11	Arterial blood gas sampling	Safely perform an arterial puncture for blood gas analysis
12	Swabs	Take microbiological swabs from a wound, the throat, or the nose for culture
13	IV infusion setup	Prepare and set up an IV infusion, including priming the line and calculating the infusion rate

Airway management and oxygen therapy

14	Oxygen therapy	Choose and apply the right oxygen device and flow rate for the clinical picture
15	Basic airway management	Perform head-tilt and chin-lift, jaw thrust, use of oropharyngeal and nasopharyngeal airways, and bag-mask ventilation

16	Advanced airway management	Perform or assist with advanced airway techniques, for example supraglottic devices and intubation

Resuscitation and life support

17	Adult BLS-D	Perform adult basic life support with defibrillation (CPR and AED use) following current guidelines
18	Pediatric BLS-D	Perform pediatric basic life support with defibrillation, using technique adapted to the child
19	ACLS	perform or take part in advanced cardiac life support following current guidelines.
20	ATLS and trauma care	assess and begin treatment of a trauma patient using a structured approach.

Acute assessment and emergency response

21	ABCDE assessment	Assess an acutely unstable patient systematically using the ABCDE approach
22	Anaphylaxis management	Recognize anaphylaxis and start the correct emergency treatment

Catheterization and tube placement

23	Urinary catheterization	Place a urinary catheter with sterile technique in both male and female patients
24	Nasogastric tube	Place a nasogastric tube and check that its position is correct

Sterile technique, wound care, and minor surgery

25	Surgical scrub, gowning, and gloving	Perform a surgical hand scrub and put on a sterile gown and gloves correctly

26	Wound assessment and dressing	Assess and disinfect a wound, and choose and apply a suitable dressing
27	Simple interrupted suturing	Place simple interrupted sutures with correct technique and secure knots
28	Intradermal or continuous suturing	Perform an intradermal or a continuous (running) suture correctly
29	Suture removal	Remove sutures safely
30	Skin staples	Apply and remove skin staples correctly
31	Local anesthesia	Give local anesthetic infiltration safely

Injection techniques

32	Subcutaneous injection	Give a subcutaneous injection with the correct site, angle, and technique
33	Intramuscular injection	Give an intramuscular injection with the correct site, angle, and technique
34	Intradermal injection	Give an intradermal injection correctly, for example for skin testing

History taking and physical examination

35	Comprehensive history taking	Take a complete, structured history: chief complaint, past medical history, medications, allergies, family history, and social history
36	Cardiovascular examination	Perform a complete cardiovascular examination (inspection, palpation, auscultation) and identify the main findings
37	Respiratory examination	Perform a complete respiratory examination and identify the main findings

38	Abdominal examination	Perform a complete abdominal examination and identify the main findings
39	Neurological examination, cranial nerves	Examine all the cranial nerves systematically
40	Neurological examination, limbs	Examine tone, strength, reflexes, coordination, and sensation of the limbs
41	Mental status assessment	Perform a structured mental or cognitive status assessment
42	Breast examination	Perform a clinical breast examination
43	Digital rectal examination	Perform a digital rectal examination
44	Gynecological examination and cervical swab	Perform a gynecological examination and take a cervical swab with correct technique
45	Newborn examination	Perform a routine newborn examination and check for the most common abnormalities

Invasive diagnostic and therapeutic procedures

46	Lumbar puncture	Perform a lumbar puncture safely, with correct positioning and technique
47	Paracentesis	Perform a diagnostic or therapeutic paracentesis safely
48	Thoracentesis and chest tube placement	Perform a thoracentesis, a chest tube placement, or both
49	Arthrocentesis	Perform an arthrocentesis with sterile technique

Interpretation of clinical data and imaging

50	ECG interpretation	Interpret a 12-lead ECG systematically and identify abnormalities
51	Blood gas interpretation	Interpret arterial and venous blood gas results, including acid-base balance and oxygenation
52	Blood test interpretation	Interpret the core laboratory panels (for example blood count, renal function, liver function, CRP) in clinical context
53	Chest X-ray interpretation	Interpret a chest X-ray systematically and identify the most common abnormalities

Table 3 - List of technical and practical skills that should be attained by medical students. In yellow are items that were deemed irrelevant/less relevant for the purpose of this thesis, and were thus not included in further analysis.

Three skills, intradermal injection, paracentesis, and arthrocentesis, highlighted in the table, were excluded from further analysis. These skills had been included in the survey because they originate from a framework that was subsequently omitted from the final thesis. None of the frameworks retained for Phase 1 (GMC, PROFILES, and Tuning) assigns a reference level to them, and no international benchmark is therefore available against which their results could be compared. The remaining 50 skills are analyzed throughout this thesis.

3.3 The Survey Instrument

3.3.1 Items

The survey has 53 items, one per technical skill, grouped into the same 12 categories used in the Phase 1 synthesis (§Phase **1: Comparison of International Frameworks**). Item wording was reviewed by two attending physicians from

the emergency department before deployment, who checked each item for clinical accuracy and clarity.

3.3.2 Scale

Both groups, students and faculty, rated each item on a six-point scale, from 0 to 5. The scale is an adapted version of the entrustment and supervision scale of ten Cate et al.[23], which was described in §Entrustable **Professional Activities (EPAs)**.

Level	**Meaning**
0	Faculty: the skill is not needed at graduation. Students: the skill is unknown to me.
1	Observe and understand.
2	Perform under direct proactive supervision - the supervisor instructs the student through each step.
3	Perform under direct reactive supervision - the supervisor intervenes or provides instructions/guidance only when deemed necessary.
4	Perform under indirect supervision - the supervisor is available to provide assistance when requested.
5	Perform autonomously.

Table 4 - Adapted entrustment and supervision scale of ten Cate et al.[23], which was submitted as part of the questionnaire in this study.

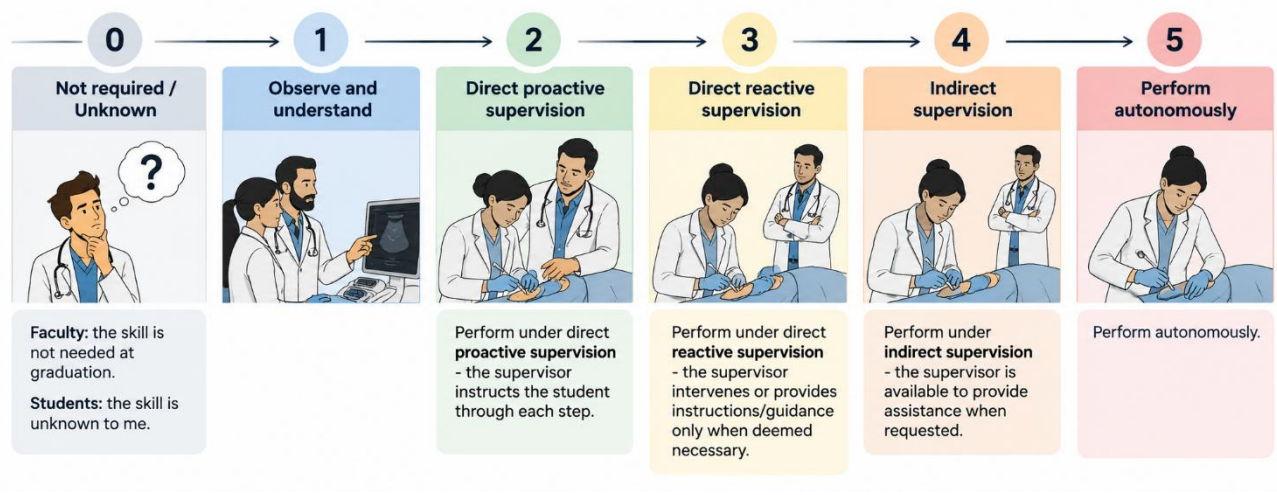


Figure 7 - Illustration and summary depicting the adjusted entrustment and supervision scale of ten Cate et al.[23]

The scales differ at level 0, since procedures not currently taught in the course may fall outside an undergraduate's knowledge entirely. A faculty rating of 0 means the skill is not required at graduation. As this represents a judgment about the target competency rather than a level of competence, it is excluded from the calculation of the faculty mean. A student rating of 0, instead, means the student has not encountered the skill at all, which is itself a real point on the competency scale as a result, it is retained and included in the student average.

3.3.3 Language

The faculty survey was administered in Italian, and the student survey in English, since Harvey is taught in English. The Italian version was produced using AI translation from the original English instrument, and it was then independently reviewed by two native Italian speakers to confirm it read naturally and fully matched the English version in meaning. Discrepancies identified during this review were resolved before the survey was distributed.

3.4 Phase 2: Faculty Survey

The faculty sample consists of Harvey course faculty who are practicing clinicians or professors, including those who retired within the past year. Participants were asked to indicate whether they also currently hold a position as department head, head of a specialty school, or faculty member without a professorial title. Participants were invited by email. The survey ran on Google Forms from 31 August 2026 to 1 October 2026. Of the 64 eligible faculty members, 26 responded, a response rate of 41%.

For each skill, the faculty level is the mean of the ratings, rounded up from a decimal value of 0.4 or above.

3.5 Phase 3: Student Self-Assessment

The student sample included all sixth-year Harvey students, which represents the closest group to the graduation reference point set in §The **Reference Point: Graduation vs. Transition to Residency**. All students were treated as one group, without separating Italian and international students. They were invited through email, class WhatsApp group, and in person during lectures; the surveys were completed on Google Forms from 15 September 2026 and 1 October 2026. Participation was anonymous and voluntary.

Students rated the level at which they consider themselves able to perform each skill now. These ratings measure perceived competence. They do not measure observed performance, and every result based on them is reported as a perceived level.

3.6 Comparison of the Three Scores and Gap Analysis

For each skill, we compared three levels: the selected international reference from Phase 1, the faculty level from Phase 2, and the student level from Phase 3. Three gaps were calculated for each skill:

- **Faculty minus student:** the main gap, between what Harvey faculty expect and what students report.
- **World minus student:** the gap between the international benchmark selected and what students report.
- **World minus faculty:** the difference between what the international frameworks expect and what Harvey faculty expect.

The first gap answers the second part of the research question; the other two show whether the course sets its own expectations lower or higher than international practice.

3.7 Phase 4: Deriving Guiding Principles and Intervention Types

Phase 4, whose objective was to propose a method to minimize or close the gap, does not produce, by itself a full curriculum; instead, the gap analysis in §Comparison **of the Three Scores and Gap Analysis** was used to derive four guiding principles for addressing the identified gaps, and, from those principles, three broad types of intervention. Skills are not prioritized by gap size alone: feasibility within Harvey's current logistical and administrative constraints is weighed alongside the size of the gap, so that a large gap on a skill with no realistic path to addressing it is not treated the same as a large gap on a skill the course can plausibly act on now. The four principles – affordability/repeatability/scalability, timing, context, and accountability – were developed by reviewing the resources and constraints described in §The **Local Context: University of Pavia**, and identifying recurring themes in where and why the largest, most addressable gaps occur. Applying these principles to the findings pointed to three concrete intervention types rather than one single curriculum change: mandatory skill components embedded within existing credit-bearing modules, short

preparatory courses timed immediately before the rotation that will implement them, and a rotation-level entrustment checklist.

This output is thematic and principle-based, not a session-by-session curriculum design, and its effect on student competence has not yet been tested.

3.8 Statistical Analysis

Both faculty and student ratings are summarized as the mean. Faculty means are rounded up from a decimal of 0.4 or above, as stated in §Phase **2: Faculty Survey**. Student means are reported without rounding. Ratings of 0 are treated differently for each group, consistent with their different meaning (§The **Survey Instrument**): faculty ratings of 0 indicate a skill is not required and are excluded from the faculty mean, whereas student ratings of 0 indicate no exposure to the skill and are retained in the student mean. The number of 0 ratings for each skill is reported separately for faculty and students.

For each skill, three gaps are calculated: Faculty minus Student, the difference between the Harvey faculty mean and the student mean; World minus Student, the difference between the international reference level (§Phase **1: Comparison of International Frameworks**) and the student mean; and World minus Faculty, the difference between the international reference level and the Harvey faculty mean. A positive value indicates the first term exceeds the second. Results are presented by the twelve skill categories defined in §Phase **1: Comparison of International Frameworks**, rather than ranked individually by gap size, to identify where the largest shortfalls concentrate for Phase 4 (§Phase **4: Deriving Guiding Principles and Intervention Types**).

This study is descriptive: no formal hypothesis is tested, and no significance threshold is applied; findings are reported as point differences rather than as statistically tested effects, consistent with the expected findings set out in §Expected **Findings**. Faculty responses are not subdivided by specialty, role, or any other characteristic; no subgroup comparison is performed.

Free-text responses to the open-ended survey questions are analyzed by content analysis: responses are analyzed for recurring themes, and illustrative quotes are reported, consistent with the approach used in §Free-Text **Responses**. This analysis is qualitative and does not produce a statistical result.

All calculations were performed in Microsoft Excel and Google Sheets.

4. Results

4.1 Response Rates and Sample Description

Twenty-six faculty members completed the survey, of 64 eligible faculty, a response rate of 40.6%. All held a professorial role: fifteen as Professore/Professoressa alone, four also holding the role of Direttore di Struttura Complessa, three also Direttore della Scuola di Specializzazione, three holding both of these roles together, and one non-tenured lecturer (Docente non di ruolo professorale).

The student survey received 144 responses across all years of the Harvey course, of which 45 were sixth-year Harvey Course students, out of a cohort of 110 sixth-year students, a response rate of 40.9%. In line with the scope set in §Scope **and Boundaries**, only these 45 responses are analyzed below; responses from other years, and the two Erasmus respondents, are not included in this section.

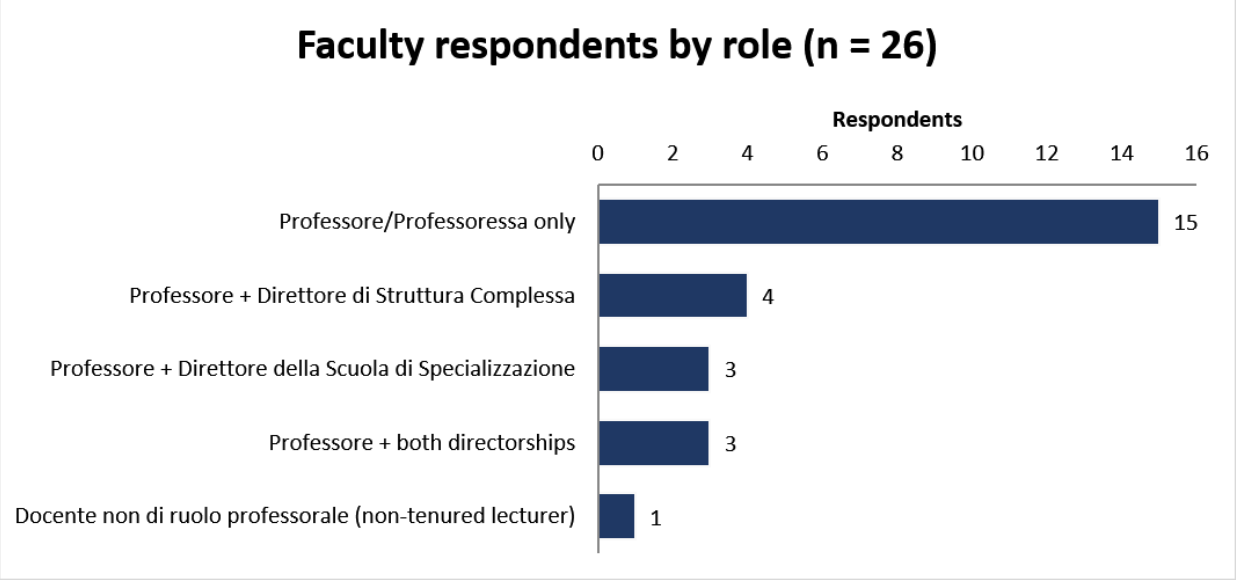


Figure 8 - Graph summarizing the faculty respondents by role for the administered questionnaire.

Group	Eligible (n)	Responded (n)	Did not respond (n)	Response rate
Faculty	64	26	38	40.6%
Sixth-year Harvey students	110	45	65	40.9%

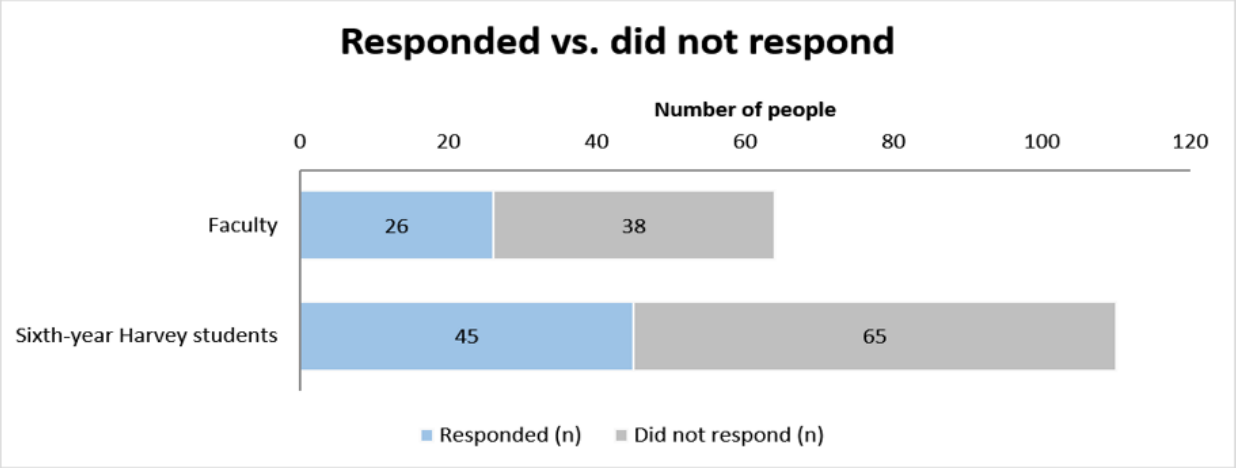


Figure 9 - Graph representation of the results of the administered questionnaire comparing faculty and student responders vs. non-responders.

4.2 International Reference Levels (Phase 1)

Of the 50 skills analyzed, 48 carry a reference level of 4 and one (skin staples) a reference level of 3, both taken from the higher of the GMC and PROFILES levels after conversion to the survey scale (§Phase **1: Comparison of International Frameworks**). One skill, thoracentesis and chest tube placement, has no reference level, because neither framework lists it. The full table is reported in Appendix A: **Phase 1 Skill Synthesis — Full List** and is not repeated here.

4.3 Faculty Expectation Results (Phase 2)

Across the 50 skills, the mean of the faculty means was 3.38. Faculty ratings of 0 ("not needed at graduation") were infrequent and are reported per skill in Appendix B: **Zero Ratings by Skill** rather than in the text; they were excluded from the faculty means, in line with §The **Survey Instrument**.

Faculty expectations reached the ceiling of the scale (mean 5, autonomous) for five skills: vital signs, comprehensive history-taking, cardiovascular examination, respiratory examination, and abdominal examination. They were lowest (mean 2) for ophthalmoscopy, spirometry, advanced airway management, ATLS/trauma care, intradermal/continuous suturing, lumbar puncture, and thoracentesis/chest tube placement.

4.4 Student Self-Assessment Results (Phase 3)

Among the 45 sixth-year respondents, the mean of the means across the 50 skills was 2.28, a full point below the equivalent faculty figure. Student ratings of 0 ("skill unknown to me") are retained in the student mean, in line with §The **Survey Instrument**, and occurred most often for ATLS/trauma care, ophthalmoscopy, and advanced airway management; the count per skill is reported in Appendix B: **Zero Ratings by Skill**.

Sixth-year students rated themselves highest for vital signs (mean 4.1) and abdominal examination (mean 3.5), and lowest for thoracentesis/chest tube placement (mean 1.1), advanced airway management and lumbar puncture (mean 1.2 each), ophthalmoscopy (mean 1.3), and nasogastric tube placement (mean 1.3).

4.5 Gap Analysis

Results are presented by the twelve skill categories defined in §Phase **1: Comparison of International Frameworks**, consistent with §Statistical **Analysis**, rather than ranked individually by gap size. Three gaps are reported for each category: ‘Faculty – Student’, ‘World – Student’, and ‘World – Faculty’, as defined in §Statistical **Analysis**.

Category	No. of skills	Mean faculty	Mean student	Mean gap (F − S)	Mean gap (W − S)	Mean gap (W − F)
11. Invasive diagnostic and therapeutic procedures	2	2	1.14	0.86	2.78	2
7. Catheterization and tube placement	2	3	1.36	1.64	2.64	1
4. Airway management and oxygen therapy	3	2.67	1.51	1.16	2.49	1.33
2. Instrumental exams and diagnostic test performance	4	2.75	2.02	0.73	1.98	1.25
3. Vascular access and sample collection	4	3.5	2.04	1.46	1.96	0.5
5. Resuscitation and life support	4	3	2.12	0.88	1.88	1
9. Injection techniques	2	4	2.27	1.73	1.73	0
6. Acute assessment and	2	3	2.34	0.66	1.66	1

emergency response						
8. Sterile technique, wound care, minor surgery	7	3.14	2.31	0.83	1.55	0.71
10. History-taking and physical examination	11	4.09	2.57	1.52	1.43	−0.09
1. Basic monitoring and point-of-care testing	5	3.8	2.87	0.93	1.13	0.2
12. Interpretation of clinical data and imaging	4	3.5	2.97	0.53	1.03	0.5

Table 5 - Summary of the evaluations given to different skill categories, including the number of skills per category, the mean value for the faculty, the mean value for the students, and the mean gaps.

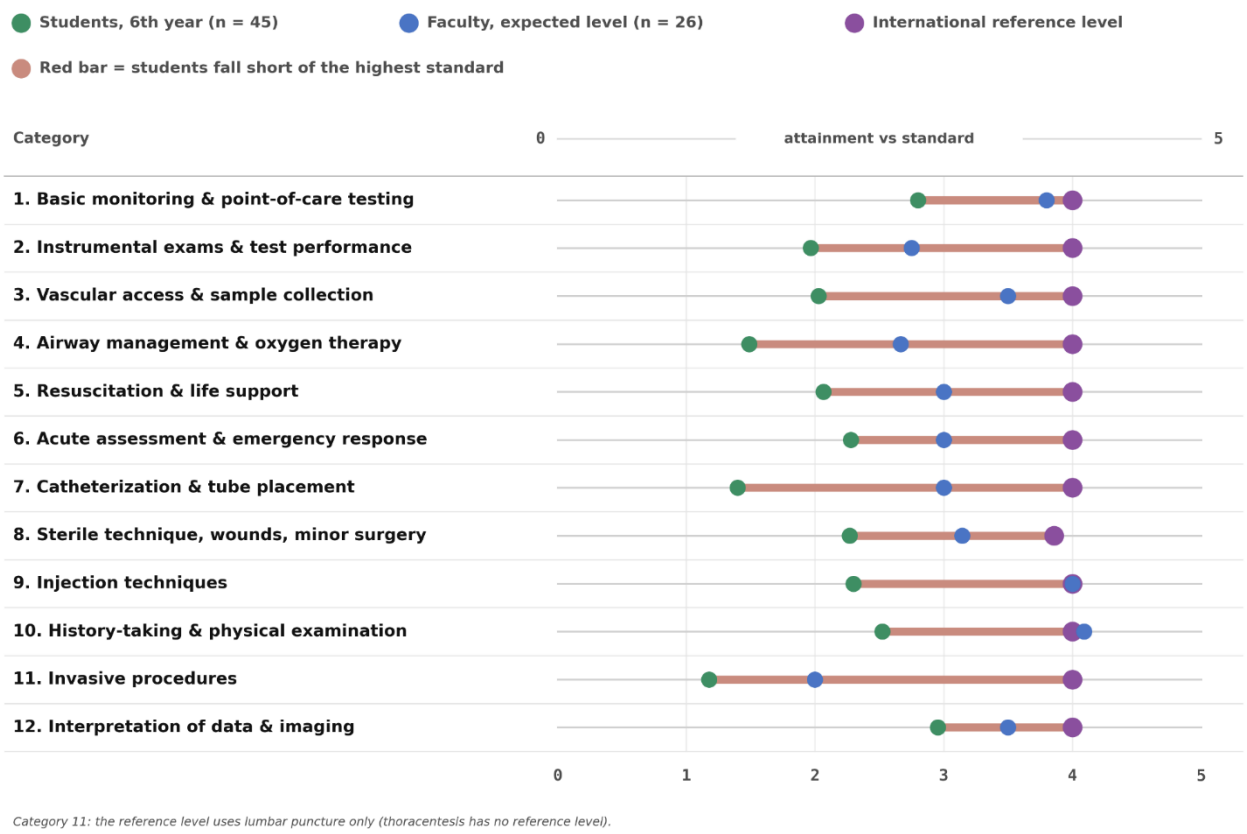


Figure 10 - Dumbbell plot summarizing the evaluations given to different skill categories.

Ranked by the 'World – Student' gap, the picture shifts from the 'Faculty – Student' ranking in a telling way: invasive procedures, catheterization, and airway management, the three categories with the least frequent, most specialized skills, now show the largest shortfalls against the international reference, each exceeding 2.4 points on average. Across the 50 skills, the mean 'World – Student' gap is 1.67, markedly larger than the mean 'Faculty – Student' gap. Thirty-nine of the 50 skills show a 'World – Student' gap of at least one point, and 19 show a gap of at least two points. The single largest individual gaps, both 2.78 points, are advanced airway management and lumbar puncture, followed closely by ophthalmoscopy, nasogastric tube placement, and ATLS/trauma care, each above 2.6 points. Students come closest to the international reference on vital signs, where their mean (4.09) slightly exceeds the reference level (4), and on abdominal

examination, 12-lead ECG, and blood test interpretation, each within 0.7 points of the reference.

Category	**No. of skills**	**Mean faculty**	**Mean student**	**Mean gap (F − S)**	**Mean gap (W − S)**	**Mean gap (W − F)**
11. Invasive diagnostic and therapeutic procedures	2	1.83	1.18	0.65	2.82	2.17
2. Instrumental exams and diagnostic test performance	4	2.46	1.97	0.49	2.03	1.54
4. Airway management and oxygen therapy	3	2.54	1.49	1.05	2.51	1.46
5. Resuscitation and life support	4	2.76	2.07	0.69	1.93	1.24
7. Catheterization and tube placement	2	2.90	1.40	1.50	2.60	1.10
6. Acute assessment and emergency response	2	3.04	2.28	0.76	1.72	0.96

8. Sterile technique, wound care, minor surgery	7	3.16	2.27	0.89	1.59	0.69
3. Vascular access and sample collection	4	3.37	2.03	1.34	1.97	0.63
12. Interpretation of clinical data and imaging	4	3.5	2.96	0.54	1.04	0.5
10. History-taking and physical examination	11	3.83	2.52	1.30	1.48	0.17
1. Basic monitoring and point-of-care testing	5	3.84	2.80	1.04	1.20	0.16
9. Injection techniques	2	4.21	2.30	1.91	1.70	−0.21

Table 6 - Summary of evaluations by skill category, including the number of skills per category, mean faculty and student scores, and mean gaps. Categories are ordered from the largest to the smallest 'World – Student' gap.

A further, distinct pattern emerges in the 'World – Faculty' column. Faculty means fall below the international reference level on 27 of the 49 referenced skills, most markedly in the invasive procedures category (mean shortfall of 2.0 points) and in airway management and oxygen therapy (mean shortfall of 1.33 points). In the history-taking and physical

examination category, however, the pattern reverses: faculty means exceed the international reference on five individual skills within it, vital signs, comprehensive history-taking, cardiovascular examination, respiratory examination, and abdominal examination, each by one full point, which pulls that category's mean ‘World – Faculty’ gap slightly negative overall (−0.09). No other category shows faculty expectations exceeding the international reference.

When analyzed together, the three gaps provide a more complete overview than any single item or gap analyzed by itself The categories with the largest ‘Faculty – Student’ gap (injection techniques, catheterization) are not the same categories with the largest ‘World – Student’ gap (invasive procedures, catheterization, airway management); catheterization appears in both, but injection techniques, the single largest ‘Faculty – Student’ gap, ranks only seventh by ‘World – Student’ gap, since faculty expectations for injections are themselves already at the international reference (‘World – Faculty’ gap of 0.00 for that category). The reverse is also true: airway management and invasive procedures show only moderate ‘Faculty – Student’ gaps, because faculty expectations are already set low there, but show the largest shortfalls against the international standard. This distinction is analyzed directly in Chapter Discussion.

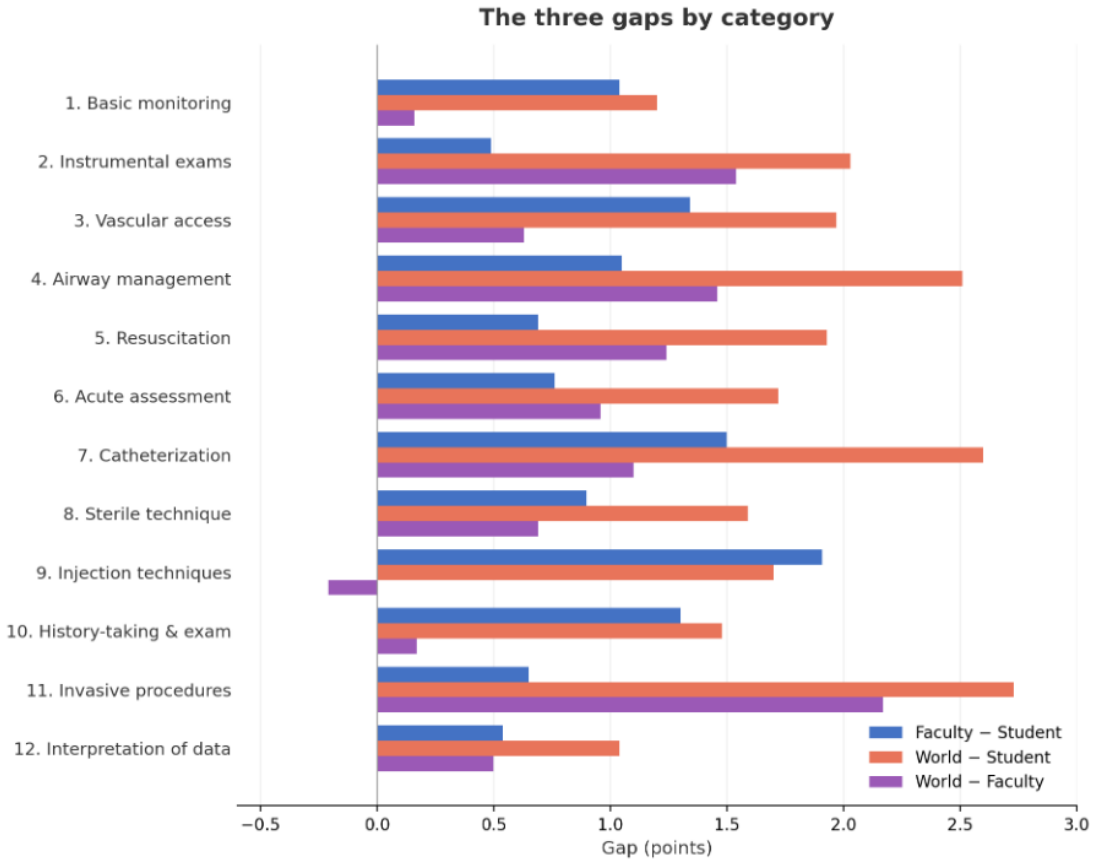


Figure 11 - Bar chart summarizing the three evaluation gaps for each skill category: the 'Faculty – Student' gap, the 'World – Student' gap, and the 'World – Faculty' gap.

4.6 Subgroup Comparisons

Faculty responses are not subdivided by specialty, role, or any other characteristic; no subgroup comparison is performed, as stated in §Statistical **Analysis**.

4.7 Free-Text Responses

Ten of the 45 sixth-year respondents answered the question on competencies missing from the list; fifteen answered the closing open comment. Several themes recurred across both:

Variability by rotation

Several respondents described competence as depending heavily on which

tirocinio, supervisor, or hospital a student was assigned to, rather than on the course itself. One wrote that "there may be huge gaps between some things," and another described acquiring most of their skills through internships abroad rather than through the course.

A perceived lack of structured practical training

Multiple respondents described the practical component as limited or inconsistent, in contrast to theoretical teaching. One wrote that "most procedures are only explained in class without any sort of practical counterpart"; another said they were "mainly a spectator in a lot of tirocini."

Specific skills named as missing or under-taught

Most often cited were basic life support (BLS-D), suturing, and point-of-care testing (blood tests and ABG). One respondent suggested formal ultrasound training; another proposed making the BLS-D course mandatory for all students.

A lack of trust extended to students on the wards

Raised as a distinct barrier to practice, not only a teaching gap. One respondent described being repeatedly asked whether they had performed a given procedure before, and, on answering no, being shown the procedure again rather than permitted to attempt it.

Concern about readiness for independent practice

Several respondents connected this directly to the stakes described in §Why **the Absence Matters More in Italy**. One wrote, "I really don't feel like I can be a doctor in few months [sic]"; another, referencing a home country's mandatory post-graduation traineeship year, said they intended to complete one abroad despite it not being required in Italy.

4.8 Summary of Key Findings

Three distinct findings emerge from the gap analysis. First, a consistent shortfall exists between what Harvey faculty expect of a graduating student and what sixth-year students report of themselves, present across most of the 50 skills analyzed and largest, at the category level, for injection technique, catheterization, and history-taking and physical examination. Second, and considerably larger in magnitude, students fall short of the international reference level by an even wider margin than they fall short of faculty expectations, most sharply for invasive procedures, catheterization, and airway management. Third, faculty expectations themselves fall below the international reference on more than half of the referenced skills, most markedly for invasive procedures and airway management, while exceeding that reference specifically on a cluster of foundational history-taking and examination skills. Taken together, the categories where students fall furthest short of the international standard are not always the categories where the course itself sets its own bar highest, a distinction with direct implications for where attention is most needed. All three findings, and their possible interpretation, are analyzed in Chapter Discussion.

5. Discussion

5.1 Principal Findings

This thesis set out to answer three questions (§Research **Question**): what level of supervised independence Harvey faculty expect a graduating student to reach in each core technical skill, how far sixth-year students' self-perceived levels fall below that expectation, and how the largest gaps could be closed within the course's existing constraints.

Three findings stand out. **First**, a faculty-student gap exists on most of the 50 skills analyzed, and it runs in one direction only: no skill showed students rating themselves at or above the level faculty expect. Twenty-nine skills showed a gap of at least one point, and the gap concentrated most heavily, at the category level, in injection techniques, catheterization and tube placement, and history-taking and physical examination. **Second**, and considerably larger, students fall short of the international reference level by a wider margin than they fall short of faculty expectations: the mean gap against the reference is 1.67 points, against a mean faculty-student gap of roughly one point, and this shortfall concentrates most heavily in invasive procedures, catheterization, and airway management. **Third**, faculty expectations themselves often sit below the international reference level built from the GMC and PROFILES frameworks in §Phase **1: Comparison of International Frameworks**. On 27 of the 49 referenced kills, more than half of the total skills analyzed, the faculty mean fell below it, in some cases by two points or more; on five skills, all core physical examination skills, the faculty mean instead exceeded the reference.

The categories driving the second and third findings are not the same categories driving the first, a discrepancy that proves more consequential than any individual numerical result reported in this section

5.2 Interpreting the Gaps

5.2.1 The 'Faculty – Student' Gap

A gap of this size and consistency is what §The **Three Pillars of Modern Medical Education** anticipated. Skill, unlike knowledge, depends on repeated, supervised, hands-on practice, and that is exactly the part of training where a widening teacher-to-student ratio strains hardest. It is also consistent with the starting argument presented in §Absence **of a National Technical Skills Consensus**: without a defined, itemized standard, students have no target to strive toward, and a student's actual competence becomes highly dependent on which rotation, ward, or supervisor they happen to be assigned to. The free-text comments in §Free-Text **Responses** illustrates this issue directly.

The pattern across skill groups adds a further detail. The categories with the largest faculty-student gap, injection techniques, catheterization, and history-taking and physical examination, share a common feature: each consists of skills a student could plausibly practice repeatedly over the course of a rotation, given the opportunity, rather than skills that are inherently rare or technically demanding. That the gap is largest here, rather than in areas requiring more complex judgment or specialized equipment, suggests this particular shortfall has less to do with a skill's difficulty than with how consistently students are actually given the chance to perform it.

5.2.2 The 'World – Student' Gap

The 'Faculty – Student' gap, despite what the previous section may suggest, is not the largest or most consistent finding in this chapter. Measured against the international reference directly, students fall short by a considerably wider margin, and the skills where that margin is widest are not the same skills driving the faculty-student gap. Invasive procedures, catheterization, and airway management show the largest 'World – Student' shortfalls, each exceeding 2.4 points on average, while injection techniques, the category with the single largest faculty-student gap, ranks only seventh against the international reference.

The underlying reason becomes apparent when the faculty perspective is incorporated into the comparison. Harvey faculty already set a comparatively ambitious local target for injection technique, meaning a student who meets that local target also comes reasonably close to the international one; the local and global pictures thus happen to align. For invasive procedures and airway management, the opposite is true: faculty expectations are themselves modest relatively to the international reference (§Faculty **Expectations vs. the International Reference**), meaning a student may fully satisfy the standards established by local faculty and nevertheless fall substantially short of the competencies expected of a graduate according to the GMC or PROFILES frameworks. The faculty-student gap, in other words, understates the true shortfall precisely where the shortfall is largest, because it is measured against a local bar that has already been set low.

This conclusion has direct implications for the appropriate target of any subsequent response. A curriculum response built only to close the faculty-student gap would, correctly, prioritize injections, catheterization, and

physical examination skills; however, it would simultaneously leave the categories with the greatest shortfall against the international standard, invasive procedures and airway management, comparatively untouched, since those categories do not appear especially urgent through the faculty-student lens alone.

5.2.3 Faculty Expectations vs. the International Reference

The reference level used throughout this thesis is, by design, the higher of the GMC and PROFILES converted levels for each skill (§Phase **1: Comparison of International Frameworks**). This is a deliberate choice, not a statistical artifact, since neither framework alone targets a fully competent, independently licensed doctor. The purpose of defining a standard at all is to produce doctors as capable as possible of entering real, unsupervised practice or residency, and a Harvey graduate, as established in §The **Reference Point: Graduation vs. Transition to Residency** and §Why **the Absence Matters More in Italy**, is legally permitted to practice independently, without supervision, from the day of graduation, with no licensing exam, no mandatory internship following graduation, and no protected training interval in which a shortfall could still be identified and corrected. In this context, the appropriate goal for most skills on this list is not a middling level of supervision, but level 4 or 5, indirect supervision or full autonomy, with lower targets reserved for skills that are, by their nature, more specialized and more reasonably left to postgraduate training: advanced or difficult airway management, ATLS-level trauma care, and comparable high-acuity or low-frequency procedures.

Measured against that goal, faculty means fall below the reference level on 27 of the 49 referenced skills. This is not a neutral or conservative finding. Rather, it represents a substantive shortcoming relative to a standard that

the course could reasonably be expected to meet. Notably, the shortfall is most pronounced in skills where it might least be anticipated: those that are common, low-risk, and non-specialized. Ophthalmoscopy and spirometry illustrate the point clearly: neither is reserved for postgraduate training, yet PROFILES expects graduates to reach indirect supervision or better on both, whereas the Harvey faculty mean is 2.0, among the lowest in the survey.

The reverse pattern is just as informative. On five skills, vital signs, comprehensive history-taking, cardiovascular examination, respiratory examination, and abdominal examination, the faculty mean reaches the ceiling of the scale and exceeds the international reference by a full point. These are the core skills of a physical examination, and they are the one item in this dataset where Harvey's own expectations are, if anything, more demanding than the international comparison. The contrast between this cluster and the shortfall described above suggests the issue is not a general reluctance among faculty to set high expectations; it is specific to certain categories, most visibly the invasive, specialized, and lower-frequency skills, and largely absent from the foundational examination skills every clinician performs routinely.

The comparison built in Phase 1 (§Phase **1: Comparison of International Frameworks**) shows that what is expected of a newly graduated doctor is broadly similar across the systems surveyed: the GMC and PROFILES converge on indirect supervision or autonomy for the great majority of technical skills. The shortfall identified in this study is thus not an indicator that the Harvey course as a whole is requiring students to meet an unusually demanding or excessive set of tasks; it is a sign that the level of expectation in Pavia itself falls short of what is otherwise a fairly consistent

international standard, concentrated in specific categories rather than spread evenly across the list. The most plausible explanation is structural rather than a matter of individual teaching quality: Harvey has no structured curriculum with explicit, year-by-year competence targets to aim for (§Absence **of a National Technical Skills Consensus**, §The **Local Context: University of Pavia**); as a result, the course has no internal mechanism forcing its expectations to track an external standard in the first place. At the same time, the kind of repeated, supervised, hands-on practice and structured clinical simulation needed to reach autonomous performance has been declining as a share of clinical teaching generally (§The **Three Pillars of Modern Medical Education**), and Pavia's own simulation infrastructure, shown to improve skills in earlier pilots (§The **Local Context: University of Pavia**), is not yet built into a required, sequenced introduction to these skills across the course. In the absence of both a defined target and a structured way of reaching it, faculty expectations have had no clear anchor to converge on, and the result is a standard that, as evidenced by the data reported in this study, sits below what the international comparison suggests a Harvey graduate should be capable of, except where years of routine clinical teaching have already produced a consistently high local expectation.

This points to a need beyond what this thesis alone can suggest. Harvey course's own stated aim is to train competent doctors for positions of responsibility, nationally and internationally (§The **Local Context: University of Pavia**), and meeting that aim requires an explicit, actionable goal-setting exercise at the course level, one this thesis's faculty survey only begins. Such an exercise should not rest on faculty opinion alone; it should draw on the Ordine dei Medici di Pavia, on clinicians working at San Matteo and the other hospitals where Harvey students train, and on recently graduated

doctors now practicing independently or in the workplaces Harvey graduates most commonly enter, so that the resulting targets more closely reflect the competencies graduates are expected to demonstrate after completing the course, while also taking account of current faculty expectations.

5.3 Comparison with Precedent

The design of this study followed Marty and colleagues' study of Swiss PROFILES graduates directly (§Entrustable **Professional Activities (EPAs)**), a comparison worth returning to. Their study found a measurable gap between Switzerland's national expected level and graduates' self-reported level across 46 EPAs. The present study finds a broadly similar pattern in the Harvey course, a gap present across most of the skill list, though the two studies differ enough in design, a pre-existing national catalogue compared against self-assessment in Switzerland, versus a locally built faculty expectation compared against self-assessment here, that a direct numerical comparison would overstate what can be concluded.

A closer parallel, in some ways, is a study from Zagreb, where graduating students' self-assessed competence across 99 clinical skills was compared with the level their own teachers expected of them, and against a separate criterion standard[57]. That study found students rated themselves consistently lower than their teachers expected, the same direction of gap found here. What this thesis adds to the literature is: a context, Italy, where no prior standard existed for the gap to be measured against at all (§Absence **of a National Technical Skills Consensus**); a course, Harvey, where the stakes of that gap are sharpened by the absence of any supervised interval after graduation (§Why **the Absence Matters More in Italy**); and a second comparison, against an international reference framework rather

than solely against local faculty expectations, which was not included in either the Swiss or Zagreb designs, and which this thesis finds to reveal the greater discrepancy.

A separate and more cautionary parallel comes from a Zambian study comparing final-year students' self-perceived competence in practical procedures against their objectively measured performance on a subsequent OSCE[56]. That study found a negative correlation between the two, students who rated themselves highly competent were, on objective assessment, found only barely competent, and concluded that self-assessment alone cannot be relied on to confirm actual readiness. This is not a comparison of findings so much as a warning about what this thesis's own findings can and cannot support, taken up directly in §Limitations **of the Study**.

5.4 From Interpretation to Proposal

When considered together, these findings point to three distinct problems rather than one:

1. **A training gap.** On most skills, and especially injection technique, catheterization, and history-taking and physical examination, students are not yet reaching what Harvey's own faculty expect of them.
2. **A readiness gap, larger than the first.** Measured against the international reference directly, students fall furthest short in invasive procedures, catheterization, and airway management, categories the training gap alone would not have flagged as the most urgent.
3. **An expectations gap.** On a majority of referenced skills, what Harvey's faculty expect is itself more conservative than at least one

comparable international framework, concentrated in the same invasive procedures and airway management categories.

These three gaps do not all pull in the same direction when it comes to building a response. Catheterization shows a large readiness gap but are, on inspection, affordable to address with resources Harvey already has. Invasive procedures and airway management are the categories where the readiness gap is just as large and the fix is genuinely harder: advanced airway techniques and invasive procedures need specialized equipment and closer supervision, and are too infrequent to practice repeatedly the way an injection or a physical examination can.

The proposal that follows holds both in view rather than resolving the tension by ignoring one side of it. It prioritizes catheterization and injection technique, where a response is both large in effect and affordable now. It treats invasive procedures and airway management as the priority for the broader goal-setting exercise called for in §Faculty **Expectations vs. the International Reference**, and for future, better-resourced intervention, rather than for an immediate fix. And where faculty expectations diverge most sharply from the international reference, it flags that divergence explicitly rather than silently building toward it.

What follows is a thematic, principle-guided proposal, built within Harvey's current logistical and administrative constraints. It is not a full, session-by-session curriculum design, and its effect on student competence has not been tested. It may nonetheless be a useful step toward understanding, and eventually addressing, the issues this thesis has identified.

5.5 Guiding Principles and Worked Examples

Principle 1: Affordability, repeatability, and scalability. Any proposed intervention must be deliverable within Harvey's current logistical and administrative constraints for the majority of skills on this list, and especially for foundational skills and for skills whose target level is 5. A proposal that cannot be repeated across an entire cohort, year after year, without new funding, new staff, or new facilities is not a realistic answer to a gap this broad.

Worked example: injection technique and catheterization, the two categories with the largest faculty-student gap, are also the most affordable to address at scale, since they need no new equipment or specialist supervision beyond what the existing Clinical Skills Lab and the 3D-printed trainers already developed with Lab3D4Med provide (§The **Local Context: University of Pavia**). Catheterization is a useful case in particular, since it shows a large gap by both the faculty-student and the world-student measure, making it one of the few categories where an affordable intervention also addresses the larger readiness gap directly, not only the local training gap. A repeatable, scheduled practice session for the whole cohort, rather than an occasional, opportunistic encounter on the wards, is achievable here without additional resources. Invasive procedures and airway management, where the readiness gap is largest, do not meet this affordability bar as easily, and are addressed separately in §Faculty **Expectations vs. the International Reference** and §Future **Work** rather than claimed as a quick win here.

Principle 2: Timing. A didactic intervention should match the knowledge students already hold at that point in the course, and should either be

repeated regularly or, where only a limited number of sessions are logistically possible, be timed for the greatest effect

Worked example 1: the Harvey study plan already offers "How to Approach an Italian Patient" as a first- and second-year elective alongside Basic Life Support and Wound Management[58]. The communication and basic physical approach content is well placed this early and, given its foundational nature, would plausibly have more impact if made compulsory, rather than elective. The repeated second-year installment, together with the practical semiotics training that students already organize for themselves on a peer-to-peer basis in the third year, shows this type of system already working informally at Harvey. It therefore requires reinforcement and further formalization rather than reinvention.

Worked example 2: Wound Management, which covers suturing technique, sits in the same first-year elective bracket. Since first-year students have essentially no clinical rotation exposure yet, a skill taught this early is unlikely to be used again for several years. Moving it later, to a point such as the fourth or fifth year where students are already on clinical traineeships and can apply the technique soon after learning it, would plausibly improve retention over teaching it in year one and not revisiting it until graduation.

Principle 3: Context. Skills should be taught paired with the courses most relevant to them, reinforcing the underlying knowledge and giving students a proper setting in which to apply the skill. This provides a genuine transition from knows to shows on Miller's pyramid (§Miller's **Pyramid**), rather than leaving procedural teaching and theoretical teaching as two disconnected tracks.

Worked example: lumbar puncture, one of the invasive procedures showing the single largest shortfall against the international reference, could be taught alongside the relevant anatomy and neurology teaching, or as part of intensive care and anesthesia, rather than as a standalone procedural session, following the same integrated design shown to improve retention of a procedural skill when paired with its underlying rationale[31]. Context-based teaching of this kind is one way a readiness-gap skill can be addressed without the full equipment or staffing investment a dedicated simulation program would require, since it uses teaching time and expertise the relevant specialty course already provides.

Principle 4: Accountability and mandatory participation. Workshops and practical sessions should be mandatory, on a predetermined schedule, allowing every student to meet a minimum participation requirement rather than leaving exposure to chance. This responds directly to the pattern described in §The **'Faculty – Student' Gap** and in students' own comments (§Free-Text **Responses**), where competence was repeatedly described as depending on which rotation or supervisor a student happened to draw.

Worked example: for the skills identified in Principle 1 as both high-gap and low-cost to address, attendance at a fixed number of scheduled practice sessions, logged per student, would ensure the gap is closed for the whole cohort rather than only for the students who happened to find or seek out the opportunity themselves.

5.6 Three Types of Intervention

The above-mentioned four principles describe how an intervention could be designed, rather than what it should comprise. Taken together, they point to three concrete types of intervention.

1. **An end-of-course workshop, covering the skills most relevant to that course.** Each course would close with a short, mandatory practical session on its own core skills, rather than leaving students to pick these up on the wards by chance. This is where the skills with the largest and most affordable gaps, injection technique and catheterization, fit most naturally: a course that already teaches the relevant theory is also the right place to require and check the matching practical skill.
2. **A rotation-prep course at the start of the year, mostly in years 3 and 4, built around that year's tirocini.** Before students start a new set of rotations, a short workshop would cover the skills those rotations would actually require, meaning what is taught is used within days or weeks rather than sitting unused for a year or more. This is where invasive procedures and airway management, the categories this proposal otherwise struggles to address affordably, are best placed: a focused session just before the rotations where they come up, rather than a standalone course with no immediate use.
3. **A rotation-level entrustment checklist, initially non-mandatory.** A short checklist, filled in at the start of a rotation, would record a student's current level for the relevant skills, using the same scale already defined in §The **Survey Instrument**. This gives the student a clear basis for asking to practice a skill, and gives the supervising

team a clear expectation that the chance should be given. It is, in effect, the entrustment logic from §Entrustable **Professional Activities (EPAs)** used at the point of care instead of only in a survey. Because it needs no new equipment or staff, it is the one intervention that fits every category in this chapter, including airway management, and it speaks directly to the lack-of-trust theme raised in §Free-Text **Responses**. Keeping it non-mandatory at first lets it be tested before any decision is made to require it.

These three types of intervention are not mutually exclusive, and a single skill could reasonably be addressed by more than one. Sequencing all three throughout the six years of the course, ensuring that each intervention lands in the year where it would have the most effect, is a natural next step once Pavia's own capacity, beyond what this thesis has established, is confirmed; confirmed; it is therefore noted here as a potential option rather than developed further, as it depends on information not yet available to this thesis.

5.7 What the Proposal Does Not Address

Three skills, intradermal injection, paracentesis, and arthrocentesis, were excluded from analysis entirely (§Phase **1: Comparison of International Frameworks**, §International **Reference Levels (Phase 1)**) and are not addressed by this proposal. A fourth skill, thoracentesis and chest tube placement, remains in the analysis but has no international reference level, since the frameworks used in this thesis do not list them; with no benchmark to measure against for this skill, there is no basis yet for deciding how it should be prioritized relative to the others.

The skills showing no measurable faculty-student gap are not excluded in the same way. Principles 2 and 3, timing and context, concern where and when a skill sits in the syllabus, not only how large its current gap is; as a result, these skills still need a defined, deliberate place in the course even where students already meet the faculty standard for them; the difference is that they are not a priority for Principle 1's limited resources.

The proposal does not address communication, clinical reasoning, or attitude, which remain outside the scope set in §Defining **"Skill": Scope of This Thesis** and §Scope **and Boundaries**, and it does not propose new resources beyond what Pavia already has in the Clinical Skills Lab, Lab3D4Med, and its existing tutoring and peer structures. For the same reason, it does not fully resolve the readiness gap identified in §The **'World – Student' Gap**, which concentrates in categories that are not, on current resources, affordable to address at the same scale as the training-gap categories.

5.8 Strengths of the Study

This is, as far as this thesis has established, the first attempt to define a technical skill standard for a Harvey graduate at all, where none existed nationally or locally beforehand (§Absence **of a National Technical Skills Consensus**). It benchmarks that standard against three international frameworks rather than one, it follows a design directly comparable to a published precedent in a comparable system, extending that design to a setting where no pre-existing national catalogue was available to measure against, and it distinguishes, where the precedent studies did not, between a local training gap and a gap against the international standard directly, a distinction this chapter has shown to carry real consequences for where attention is most needed.

5.9 Limitations of the Study

The central limitation is that both the faculty and student scores in this study are self-reported, not observed. Self-assessed competence has been shown to correlate poorly, and in at least one comparable study negatively, with objectively measured performance[55,56]. The gaps reported here should therefore be read as gaps in perceived competence and perceived expectation, not confirmed gaps in actual clinical performance.

Faculty expectation was established through a single survey round rather than a full Delphi process with iterative feedback, from 26 of 64 eligible faculty (40.6%). The student sample, 45 of 110 eligible sixth-year students (40.9%), is similarly a convenience sample rather than a full census. The reference level itself is an upper bound, the highest level across the GMC and PROFILES, not an average, which means faculty-vs-reference and student-vs-reference comparisons in particular should be read as conservative rather than inflated, but also means they cannot distinguish an expectation that is slightly below a demanding but reasonable target from one that is well below every framework consulted. Three skills were excluded from analysis entirely, and a fourth has no reference level at all (§What **the Proposal Does Not Address**). The findings describe the Harvey course specifically and do not extend to Golgi (§Scope **and Boundaries**). No subgroup comparison was performed (§Statistical **Analysis**); faculty responses were not subdivided. Finally, the intervention types proposed in §Three **Types of Intervention** have not been piloted as a complete, coordinated program; related approaches have been tried in Pavia only as fragmented, stand-alone efforts, and their effect on student competence remains untested.

5.10 Implications

For Pavia, these findings suggest two distinct priorities rather than one. The training-gap categories, injection technique and catheterization in particular, are where an affordable intervention could show measurable improvement quickly, using resources already in place. The readiness-gap categories, invasive procedures and airway management, are where the shortfall against international practice is largest, but where closing it will likely require resources beyond what this thesis has been able to establish as currently available; these categories are better treated as a priority for the broader goal-setting exercise called for in §Faculty **Expectations vs. the International Reference** than as a target for immediate, low-cost action.

For Italy more broadly, the same structural gap identified in §Absence **of a National Technical Skills Consensus** is visible in this thesis in concrete form. A course whose own faculty do not converge on a single, internationally consistent expectation, and whose students still fall short against that external standard, is a course operating without the kind of national reference point that the GMC or PROFILES already provide elsewhere. This thesis offers one case study of what measuring that absence, even partially and provisionally, can reveal, and suggests the same exercise would likely reveal comparable gaps wherever it is repeated.

5.11 Future Work

The most immediate next step is the broader goal-setting exercise called for in §Faculty **Expectations vs. the International Reference**: a dedicated process, beyond this thesis's faculty survey, to set explicit, actionable, year-by-year competence targets for the Harvey course, with input from the Ordine dei Medici di Pavia, clinicians at San Matteo and the other teaching hospitals, and recently graduated doctors now in independent practice.

Given §The **‘World – Student’ Gap**'s finding, this exercise should pay particular attention to the readiness-gap categories, invasive procedures and airway management, where the shortfall against international practice is largest but where this thesis's own feasibility constraint has limited what could be proposed.

The data collection itself should also be extended before any intervention is finalized. The current faculty and student samples, while substantial, remain convenience samples rather than a full census of either population; both should be expanded toward complete coverage, with the student survey ideally timed toward February, when sixth-year formal teaching concludes, so that self-assessment reflects as much of the course as possible. The same design should also be extended to Golgi, as the pilot nature of this study anticipated from the outset (§The **Local Context: University of Pavia**), to test whether these findings are specific to Harvey's cohort and structure. Beyond faculty and students, the rater base itself should widen to include more attendings across the hospitals where students rotate and recently graduated doctors now working independently, so that the standard is triangulated across more than one perspective within the course.

Once a larger dataset and the broader goal-setting exercise are both in place, the next step is translating §Three **Types of Intervention**'s three intervention types into an actual, implementable list of workshops and short courses, embedded within Harvey's existing modules rather than added alongside them, and piloting that list on a limited scale to measure whether both the training gap and the readiness gap narrow. A full Delphi process, rather than a single survey round, would strengthen the faculty expectation underlying all of this. Finally, validating a sample of self-assessed levels against directly observed performance, an OSCE station or a structured

workplace assessment, would address this study's central limitation directly.

6. Conclusions

This thesis set out to define, for the first time, a technical clinical skill standard for a graduating Harvey student, to measure how far current students stand from it, and to propose a way of closing the distance.

On the first question, Harvey faculty, once surveyed directly, produced a usable expectation standard across 50 technical skills, one that exceeds the international reference on a cluster of foundational examination skills but falls below it on more than half of all referenced skills, most markedly for invasive and specialized procedures. On the second, sixth-year students' self-perceived competence falls short of both the faculty standard and, by a wider margin, the international reference itself, most sharply for invasive procedures, catheterization, and airway management. On the third, this thesis proposes four guiding principles and three types of intervention, prioritizing the categories where a training-gap response is both large and immediately affordable, while identifying the categories driving the larger readiness gap as the priority for the broader goal-setting exercise this thesis recommends but could not itself complete.

Where none existed before, this thesis provides the University of Pavia's Harvey course with its first internationally benchmarked, faculty-endorsed technical skill standard, a provisional account of how far the course currently stands from it, and a distinction, between a local training gap and a gap against international practice, that any future effort to close it should keep in view.

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Appendix

Appendix A: Phase 1 Skill Synthesis — Full List

This appendix gives the complete list of 53 technical clinical skills identified in Phase 1 (§Phase **1: Comparison of International Frameworks**), with each framework's coverage and the resulting reference level used in Chapters Results and Discussion.

Legend

GMC/PROFILES columns. The level (0 to 4) at which each framework expects the skill to be performed at graduation, after conversion to the 0 to 5 supervision scale defined in §The **Survey Instrument**. A value of 0 means the framework does not list the skill, or lists it at a level that converts to 0.

Tuning column. Tuning gives no supervision levels (§Phase **1: Comparison of International Frameworks**), so this column instead records coverage: **Y** = the skill is listed, **N** = not listed, **P** = partially covered, the skill overlaps with a broader Tuning item rather than being named directly.

Reference level. The reference level is the higher of the GMC and PROFILES values for each skill. Tuning is not counted toward the reference level, since it supplies no level, only coverage (§Phase **1: Comparison of International Frameworks**). Where neither GMC nor PROFILES lists a skill, the skill has no reference level.

Skills excluded from the Phase 2/3 gap analysis. Three skills in this list, intradermal injection (34), paracentesis (47), and arthrocentesis (49), were excluded from the analysis in Chapters Results and Discussion (§Phase **1: Comparison of International Frameworks**, §International **Reference Levels (Phase 1)**). They are included here for completeness, since they remain part of the full Phase 1 synthesis, but their gap figures do not appear elsewhere in this

thesis. Thoracentesis and chest tube placement (48) was not excluded and remains part of the 50-skill analysis.

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1. Basic monitoring and point-of-care testing

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level
1	Vital signs	Measure and record blood pressure, heart rate, respiratory rate, temperature, and oxygen saturation, and tell normal from abnormal values	4	4	P	4
2	Capillary blood glucose	Perform a fingerstick glucose test and interpret the result	4	0	N	4
3	Urinalysis, urine dipstick,	Perform and read a urine dipstick and	4	4	Y	4

	pregnancy test	a urinary pregnancy test, and interpret the results				
4	Glasgow Coma Scale and ACVPU	Assess and score level of consciousness with the GCS and ACVPU in an acute setting	0	4	N	4
5	NEWS2 and early warning scores	Calculate an early warning score from vital signs and use it to recognize a deteriorating patient	0	4	N	4

2. Instrumental exams and diagnostic test performance (performing the test, not interpreting it)

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level

6	Ophthalmoscopy	Use a direct ophthalmoscope to see the fundus and identify obvious abnormalities	4	4	N	4
7	Otoscopy	Use an otoscope to examine the external ear canal and the tympanic membrane	4	4	N	4
8	12-lead ECG	Place the electrodes correctly and record a technically adequate 12-lead ECG	4	4	N	4
9	Spirometry	Perform basic spirometry and guide the patient	4	4	Y	4

		through the maneuver				

3. Vascular access and sample collection

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level
10	Venous access	Perform a venous blood draw, place a peripheral IV cannula, and obtain a blood culture with correct technique	3-4	4	P	4
11	Arterial blood gas sampling	Safely perform an arterial puncture for blood gas analysis	3	4	N	4
12	Swabs	Take microbiological swabs from a wound, the throat, or the nose for culture	4	4	N	4

13	IV infusion setup	Prepare and set up an IV infusion, including priming the line and calculating the infusion rate	3	4	Y	4

4. Airway management and oxygen therapy

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level
14	Oxygen therapy	Choose and apply the right oxygen device and flow rate for the clinical picture	4	0	Y	4
15	Basic airway management	Perform head-tilt and chin-lift, jaw thrust, use of oropharyngeal and nasopharyngeal airways,	0	4	N	4

		and bag-mask ventilation				
16	Advanced airway management	Perform or assist with advanced airway techniques, for example supraglottic devices and intubation	0	4	P	4

5. Resuscitation and life support

No.	**Skill**	**What it means**	**GMC**	**PROFILES**	**Tuning**	**Reference level**
17	Adult BLS-D	Perform adult basic life support with defibrillation (CPR and AED use) following current guidelines	0	4	N	4
18	Pediatric BLS-D	Perform pediatric basic life	0	4	N	4

		support with defibrillation, using technique adapted to the child				
19	ACLS	Perform or take part in advanced cardiac life support following current guidelines	0	4	Y	4
20	ATLS and trauma care	Assess and begin treatment of a trauma patient using a structured approach	0	4	Y	4

6. Acute assessment and emergency response

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level
21	ABCDE assessment	Assess an acutely unstable	0	4	P	4

		patient systematically using the ABCDE approach				
22	Anaphylaxis management	Recognize anaphylaxis and start the correct emergency treatment	0	4	N	4

7. Catheterization and tube placement

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level
23	Urinary catheterization	Place a urinary catheter with sterile technique in both male and female patients	3	4	Y	4
24	Nasogastric tube	Place a nasogastric tube and	2	4	N	4

		check that its position is correct				

8. Sterile technique, wound care, and minor surgery

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level
25	Surgical scrub, gowning, and gloving	Perform a surgical hand scrub and put on a sterile gown and gloves correctly	3	4	N	4
26	Wound assessment and dressing	Assess and disinfect a wound, and choose and apply a suitable dressing	3	4	N	4
27	Simple interrupted suturing	Place simple interrupted sutures with	3	4	Y	4

		correct technique and secure knots				
28	Intradermal or continuous suturing	Perform an intradermal or a continuous (running) suture correctly	3	4	P	4
29	Suture removal	Remove sutures safely	0	4	N	4
30	Skin staples	Apply and remove skin staples correctly	3	0	N	3
31	Local anesthesia	Give local anesthetic infiltration safely	3	4	N	4

9. Injection techniques

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level

32	Subcutaneous injection	Give a subcutaneous injection with the correct site, angle, and technique	3	4	N	4
33	Intramuscular injection	Give an intramuscular injection with the correct site, angle, and technique	3	4	N	4
34	Intradermal injection *(excluded from analysis)*	Give an intradermal injection correctly, for example for skin testing	0	0	N	None

10. History-taking and physical examination

No.	**Skill**	**What it means**	**GMC**	**PROFILES**	**Tuning**	**Reference level**
35	Comprehensive history taking	Take a complete, structured	0	4	P	4

		history: chief complaint, past medical history, medications, allergies, family history, and social history				
36	Cardiovascular examination	Perform a complete cardiovascular examination (inspection, palpation, auscultation) and identify the main findings	0	4	N	4
37	Respiratory examination	Perform a complete respiratory examination	0	4	N	4

		and identify the main findings				
38	Abdominal examination	Perform a complete abdominal examination and identify the main findings	0	4	N	4
39	Neurological examination, cranial nerves	Examine all the cranial nerves systematically	0	4	N	4
40	Neurological examination, limbs	Examine tone, strength, reflexes, coordination, and sensation of the limbs	0	4	N	4
41	Mental status assessment	Perform a structured mental or cognitive	0	4	N	4

		status assessment				
42	Breast examination	Perform a clinical breast examination	0	4	N	4
43	Digital rectal examination	Perform a digital rectal examination	0	4	N	4
44	Gynecological examination and cervical swab	Perform a gynecological examination and take a cervical swab with correct technique	0	4	N	4
45	Newborn examination	Perform a routine newborn examination and check for the most common abnormalities	0	4	N	4

11. Invasive diagnostic and therapeutic procedures

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level
46	Lumbar puncture	Perform a lumbar puncture safely, with correct positioning and technique	0	4	N	4
47	Paracentesis *(excluded from analysis)*	Perform a diagnostic or therapeutic paracentesis safely	0	0	N	None
48	Thoracentesis and chest tube placement	Perform a thoracentesis, a chest tube placement, or both	0	0	N	None
49	Arthrocentesis *(excluded from analysis)*	Perform an arthrocentesis with	0	0	N	None

		sterile technique				

12. Interpretation of clinical data and imaging

No.	Skill	What it means	GMC	PROFILES	Tuning	Reference level
50	ECG interpretation	Interpret a 12-lead ECG systematically and identify abnormalities	4	4	N	4
51	Blood gas interpretation	Interpret arterial and venous blood gas results, including acid-base balance and oxygenation	3	4	N	4
52	Blood test interpretation	Interpret the core laboratory panels (for example	0	4	N	4

		blood count, renal function, liver function, CRP) in clinical context				
53	Chest X-ray interpretation	Interpret a chest X-ray systematically and identify the most common abnormalities	0	4	N	4

Appendix B: Zero Ratings by Skill

Counts of faculty ratings of 0 ("not needed at graduation") and student ratings of 0 ("skill unknown to me") for each of the 50 skills analyzed in Chapter Results. Faculty 0 ratings are excluded from the faculty mean; student 0 ratings are retained in the student mean (§The **Survey Instrument**, §Statistical **Analysis**). n = 26 faculty respondents, 45 sixth-year student respondents, for every skill.

No.	Skill	Faculty rated 0	Faculty rated >0	Student rated 0	Student rated >0
1	Vital signs	0	26	1	45
2	Capillary blood glucose	0	26	2	45
3	Urinalysis/dipstick/pregnancy test	0	26	3	45
4	GCS + ACVPU	0	26	0	45
5	NEWS2/early warning score	1	25	9	45
6	Ophthalmoscopy	5	21	11	45
7	Otoscopy	1	25	5	45
8	12-lead ECG	1	25	2	45
9	Spirometry	2	24	5	45
10	Venous access (draw/cannula/culture)	0	26	4	45
11	Arterial blood gas sampling	0	26	4	45
12	Swabs	0	26	3	45
13	IV infusion setup	1	25	8	45
14	Oxygen therapy	0	26	7	45
15	Basic airway management	0	26	6	45
16	Advanced airway management	4	22	11	45
17	Adult BLS-D	0	26	1	45
18	Pediatric BLS-D	1	25	7	45
19	ACLS	2	24	10	45
20	ATLS/Trauma care	2	24	13	45
21	ABCDE assessment	0	26	1	45

22	Anaphylaxis management	1	25	3	45
23	Urinary catheterization	0	26	8	45
24	Nasogastric tube	0	26	7	45
25	Surgical scrub/gown/gloves	0	26	4	45
26	Wound assessment and dressing	0	26	4	45
27	Simple interrupted suturing	1	25	6	45
28	Intradermal/continuous suturing	1	25	9	45
29	Suture removal	0	26	5	45
30	Skin staples	1	25	7	45
31	Local anesthesia	0	26	8	45
32	Subcutaneous injection	0	26	4	45
33	Intramuscular injection	0	26	2	45
35	Comprehensive history-taking	0	26	0	45
36	Cardiovascular exam	0	26	0	45
37	Respiratory exam	0	26	0	45
38	Abdominal exam	0	26	0	45
39	Neuro exam - cranial nerves	0	26	0	45
40	Neuro exam - limbs	0	26	0	45
41	Mental status assessment	0	26	1	45
42	Breast exam	0	26	5	45
43	Digital rectal exam	0	26	8	45
44	Gynecological exam/cervical swab	1	25	7	45
45	Newborn exam	0	26	10	45

46	Lumbar puncture	1	25	5	45
48	Thoracentesis/chest tube	0	26	9	45
50	ECG interpretation	0	26	4	45
51	Blood gas interpretation	0	26	1	45
52	Blood test interpretation	0	26	1	45
53	Chest X-ray interpretation	0	26	2	45

Totals. Faculty rated a skill 0 on 26 occasions across all 50 skills, out of a possible 1300. Students rated a skill 0 on 233 occasions across all 50 skills, out of a possible 2250.

The skills most often rated 0 by faculty were ophthalmoscopy (5), advanced airway management (4), ACLS and ATLS/trauma care (2 each). The skills most often rated 0 by students were ATLS/trauma care (13), ophthalmoscopy and advanced airway management (11 each), and newborn examination (10).

Appendix C: Intra-Student Variability in Self-Rated Clinical Skill Readiness

This appendix supports the claim in §The **Problem** that, without a predefined standard, students may lack clear guidance on which skills to prioritize and to what level of independence to strive, and that this can contribute to variability in students' clinical and technical experience by graduation. It reports how much sixth-year Harvey students differ from one another in self-rated readiness, using the 50 skills analyzed in Chapter **Results**4; the three skills excluded from that analysis (§3.3, §4.2) are excluded here as well.

What is reported

For each of the 45 sixth-year respondents, an overall readiness score was calculated by averaging their self-ratings across the 50 skills. This appendix looks at how those overall scores varied between students. A wider spread indicates greater differences in self-rated readiness at the end of the course, while a narrower spread suggests that students reached a more similar level.

Two statistics summarize this spread:

- **Standard deviation (SD).** How far, on average, a student's overall readiness score sits from the group mean.
- **Coefficient of variation (CV).** The SD expressed as a percentage of the group mean, which makes the spread easier to interpret in proportion to the scale on which it is measured.

Overall readiness across students

Statistic	Value
Mean	2.25
Median	2.22
Standard deviation (SD)	0.81
Coefficient of variation (CV)	36.1%
Minimum	0.36
Maximum	4.08

Range	3.72

Across the 45 respondents, overall readiness ranged from 0.36 to 4.08 on the 0 to 5 scale, a difference of 3.72 points. A CV of 36.1% indicates that the spread between students is large relative to the average score itself.

Variability by skill category

The same comparison, run separately for each of the 12 skill categories defined in §3.3, shows that this spread is not uniform across the skill list.

Category	**No. of skills**	**Mean**	**SD**	**CV (%)**
1. Basic monitoring and point-of-care testing	5	2.80	1.07	38.1
2. Instrumental exams and diagnostic test performance	4	1.97	0.88	44.7
3. Vascular access and sample collection	4	2.03	1.11	54.8
4. Airway management and oxygen therapy	3	1.49	1.02	68.8
5. Resuscitation and life support	4	2.07	1.37	66.1
6. Acute assessment and emergency response	2	2.28	0.99	43.3

7. Catheterization and tube placement	2	1.40	0.89	63.5
8. Sterile technique, wound care, minor surgery	7	2.27	1.24	54.6
9. Injection techniques	2	2.30	1.31	57.0
10. History-taking and physical examination	11	2.52	0.97	38.6
11. Invasive diagnostic and therapeutic procedures	2	1.18	0.69	58.8
12. Interpretation of clinical data and imaging	4	2.96	1.10	37.2

The spread between students is narrowest for history-taking and physical examination and for interpretation of clinical data and imaging (CV of 38.6% and 37.2%), and widest for airway management and resuscitation and life support (CV of 68.8% and 66.1%). Categories built around routine (frequently practiced) skills show the most consistent readiness across students; categories built around less frequent or more specialized skills show the widest disagreement between students about their own readiness.

Interpretation

Students reach the sixth year with markedly different levels of self-rated readiness, both overall and most visibly in the categories tied to less frequent clinical skills. This is consistent with the explanation offered in §2.1 and §5.2.1: without a shared, explicit target for each skill, what a

given student has been taught and allowed to practice depends on which rotation, supervisor, and hospital they were assigned to, rather than on a common standard applied across the course. As with the rest of this thesis, these are self-reported ratings of confidence, not observed competence (§5.9), so the variability reported here describes differences in self-perceived readiness, which may reflect differences in actual clinical exposure, in opportunities for supervised practice, or in how confidently different students rate themselves.